

ELECTROCHEMISTRY

Electrochemistry is the branch of physical chemistry which deals with the study of conversion of

- * chemical energy into electrical energy (Galvanic cell)
- * Electrical energy into chemical energy (Electrolytic cell)

PART - ONE - GENERAL POINTS

1. CONDUCTANCE

Those substances which can conduct electricity are called conductors. Conductors are of two types

- * Metallic Conductors (electronic conductors)
 - * Electrolytic conductors (Ionic conductors)
-] BOTH FOLLOW OHM'S LAW

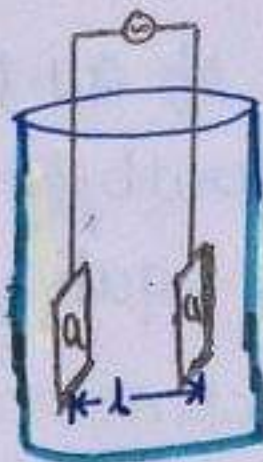
METALLIC CONDUCTOR	ELECTROLYTIC CONDUCTOR
Flow of current due to movement of free electrons	Flow of current due to movement of ions
Does not involve transfer of matters	Involves transfer of matters
Does not involve any chemical changes	Involves chemical changes
conducting power is usually very high	conducting power is comparatively less
Does not follow faraday's law of electrolysis	follows faraday's law of electrolysis
On increasing Temperature Resistance increases Conductivity decreases	On increasing Temperature Resistance decreases Conductivity increases

2. CONDUCTANCE TERMS

IN CHEMISTRY CONCEPT OF [K]

- * l = distance between electrode (generally in cm)

Battery कोई metre या दो मीटर का generally नहीं होता है।



- * A = Dipped surface area

ये surface area लेते वक्त दिमाग और आँख का बनी जला लो वरना सामने wrong answer खड़ा है।

- * l and A don't depend upon nature of electrolysis.

- * for a given cell l and A always remain same

therefore $\frac{l}{A}$ is also called cell constant $[C^*/G^*]$

Reason ये है कि Battery का size एक बार जो बन गया सो बन गया अब Battery थोड़े ना अपना length और area increase या decrease करेगा that's why $\frac{l}{A}$ is constant and called cell constant $[C^*/G^*]$

$$K = \frac{l}{R} \cdot \frac{l}{A}$$

$$K = G \times G^*$$

K = conductance $\times$ cell constant

if $l = 1\text{ cm}$
 $A = 1\text{ cm}^2$

then $K = G$ so we can say that conductance of

$1\text{ cm}^3/1\text{ M}$ solution is called as specific conductance
 or
 conductivity

2.C. EQUIVALENT CONDUCTANCE [λ_{eq} OR Λ_{eq}]

Conductance of all the ions produced by **1 gm equivalent** of an electrolyte **in given volume** of solution is called as equivalent conductance

∴ conductance of 1 cm^3 solution = K

∴ conductance of $V \text{ cm}^3$ solution = $K \times V$

$$\boxed{\lambda_{eq} = K \times V}$$

↓
vol^m of solution (ml) which contain
1 gm eq of an electrolyte

Suppose concentration is $N \text{ eq/Lit}$

∴ vol^m of solution containing $N \text{ gm eq} = 1000 \text{ ml}$

∴ vol^m of solution containing $1 \text{ gm eq} = \frac{1000}{N}$

$$\lambda_{eq} = K \times V$$

$$\boxed{\lambda_{eq} = K \times \left(\frac{1000}{N}\right)}$$

→ vol^m of solution (ml) which contain
1 gm eq of an electrolyte

$$\text{UNIT } [\lambda_{eq} / \Lambda_{eq}] = \Omega^{-1} \text{ cm}^{-1} \times \text{cm}^3 \text{ eq}^{-1}$$

$$= \Omega^{-1} \text{ cm}^2 \text{ eq}^{-1}$$

$$= \text{S cm}^2 \text{ eq}^{-1}$$

2.D- MOLAR CONDUCTANCE [λ_M OR Λ_M OR μM]

Conductance of all the ions produced by **1 mol** of an electrolyte in **given volume** of solution is called as **MOLAR CONDUCTANCE**

$$\because \text{conductance of } 1 \text{ cm}^3 \text{ solution} = K$$

$$\therefore \text{conductance of } V \text{ cm}^3 \text{ solution} = K \times V$$

$$\boxed{\lambda_M = K \times V}$$

$\downarrow$
vol^m of solution (ml) which contain **1 mol** of electrolyte

$$\because \text{volume of solution containing } M \text{ molar} = 1000 \text{ ml}$$

$$\therefore \text{ " " " " " } 1 \text{ molar} = \frac{1000}{M} \text{ ml}$$

$$\boxed{\lambda_M = K \times V}$$

$$\boxed{\lambda_M = K \times \left(\frac{1000}{M} \right)}$$

$\rightarrow$ vol^m of solution (ml) which contain **1 mol** of electrolyte

$$\begin{aligned} \text{UNIT } [\lambda_M / \Lambda_M / \mu M] &= \Omega^{-1} \text{ cm}^{-1} \times \text{cm}^3 \text{ eq}^{-1} \\ &= \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1} \\ &= \text{S cm}^2 \text{ mol}^{-1} \end{aligned}$$

RELATION BETWEEN λ_M AND λ_{eq}

$$\lambda_{eq} = K \times \frac{1000}{N}$$

$$\lambda_{eq} = K \times \frac{1000}{M \times v.f}$$

$$\lambda_{eq} = \frac{\lambda_M}{v.f}$$

$$\boxed{\lambda_{eq} \times v.f = \lambda_M}$$

Min^m value of v.f 1 तो होगा ही होगा so

$$\boxed{\lambda_M \geq \lambda_{eq}}$$

conductance and conductivity are additive property
 जैसे कि अगर कोई electrolytic solution में several electrolytic
 ions हैं तो total conductance of solution will be

$$C_{\text{TOTAL}} = \sum C_{\text{ION}} + \sum C_{\text{WATER}}$$

FORMULAE

$$1. \quad G = \frac{1}{R}$$

$$2. \quad K = \frac{1}{R} \cdot \frac{L}{A} = G \times G^* = \frac{G}{R}$$

$$3. \quad \lambda_{eq} = K \times V = K \times \frac{1000}{N}$$

$$4. \quad \lambda_{eq} = K \times V = K \times \frac{1000}{M}$$

$$5. \quad \lambda_M = \lambda_{eq} \times V \cdot f$$

Qsn The resistance of $0.005M$ K_2SO_4 at $25^\circ C$ is 200Ω . Cell constant (K) of conducting cell is 1 cm^{-1} . Then find equivalent conductance.

Ans

$$\begin{aligned}\lambda_{eq} &= K \times \frac{1000}{N} \\ &= K \times \frac{1000}{M \times v \cdot f} \\ &= \frac{1}{R} \times G^\circ \times \frac{1000}{M \times v \cdot f} \\ &= \frac{1}{200} \times (1) \times \frac{1000}{0.005 \times 2} \\ &= \underline{500 \Omega^{-1} \text{ cm}^2 \text{ eq}^{-1}}\end{aligned}$$

Qsn Resistance of Decinormal solution present between two platinum electrode that are 1.8 cm apart dipped and 5.4 cm^2 area found to be 50Ω . Then calculate λ_{eq} of solution.

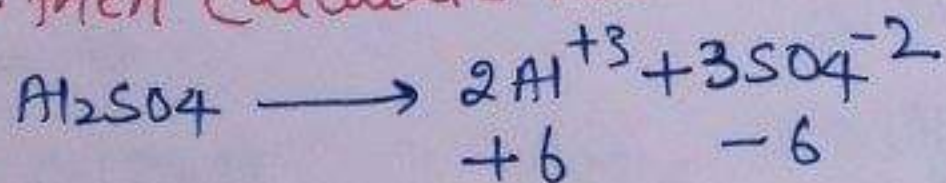
Ans

$$\begin{aligned}\lambda_{eq} &= K \times \frac{1000}{N} \\ &= \frac{1}{R} \times \frac{l}{a} \times \frac{1000}{N} \\ &= \frac{1}{50} \times \frac{1.8}{5.4} \times \frac{1000}{1/10} \\ &= \underline{66.67 \Omega^{-1} \text{ cm}^2 \text{ eq}^{-1}}\end{aligned}$$

Qsn HOME WORK

$0.5N$ $Al_2(SO_4)_3$ solution is present between two platinum electrode that are $2\text{ cm} \times 4\text{ cm}$. The resistance of solution is 200Ω . Then calculate molar conductance of solution.

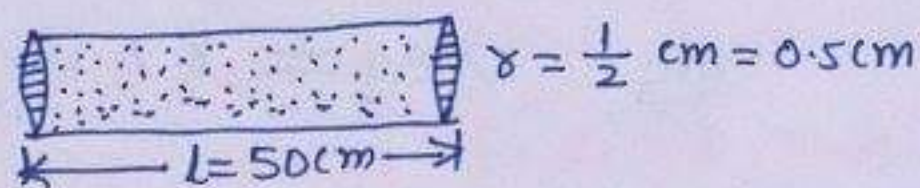
Hints



$$\lambda_M = \frac{K \times 1000 \times v \cdot f}{N}$$

Ques Resistance of a column of 0.5 mole per litre NaOH solution of diameter 1cm and length 50cm is $5.5 \times 10^3 \Omega$. calculate its Resistivity, conductivity and molar conductivity

Ans $R = \rho \cdot \frac{l}{A}$



$$\rho = \frac{RA}{l} = \frac{R(\pi r^2)}{l}$$

$$\rho = \frac{5.5 \times 10^3 \times 3.14 \times (0.5)^2}{50}$$

$$\boxed{\rho = 87.135 \Omega\text{-cm}}$$

$$k = \frac{1}{\rho} = \frac{1}{87.135 \Omega\text{-cm}} = 0.01148 \Omega^{-1}\text{cm}^{-1}$$

$$\Lambda_M = k \times \frac{1000}{M} = 0.01148 \Omega^{-1}\text{cm}^{-1} \times \frac{1000}{0.5}$$

OPTION में ये तीनों
में से कोई एक
रहेगा. जैसे
नीचे दिया है।

$$\left[\begin{aligned} \Lambda_M &= 229.6 \Omega^{-1}(\text{cm}^2) \text{mol}^{-1} \\ &= 229.6 \Omega^{-1}(10^{-4} \text{m}^2) \text{mol}^{-1} \\ &= 229.6 \times 10^{-4} \Omega^{-1} \text{m}^2 \text{mol}^{-1} \end{aligned} \right.$$

(i) $\Lambda_M = 229.6 \Omega^{-1}(\text{cm}^2) \text{mol}^{-1}$

(ii) $\Lambda_M = 229.6 \Omega^{-1}(10^{-4} \text{m}^2) \text{mol}^{-1}$

(iii) $\Lambda_M = 229.6 \times 10^{-4} \Omega^{-1} \text{m}^2 \text{mol}^{-1}$

(iv) All are correct

Ans = (iv)

Qsn Equivalent conductance of 0.1M solution of $MgCl_2$ is $100 \text{ mho cm}^2 \text{ eq}^{-1}$ at 25°C . A cell with electrode that are 1.5 cm is surface area and 0.5 cm apart is filled with 0.1N $MgCl_2$ solution. How much current will flow when the potential difference between electrode is 5V.

Ans $V = IR$ $I = \frac{V}{R}$ [V तो है But R find करना है]

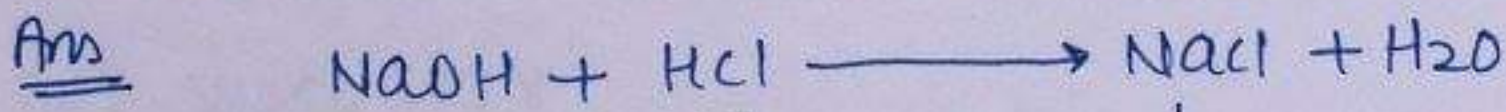
$$\Lambda_{eq} = K \times \frac{1000}{N}$$

$$100 = \frac{1}{R} \times \frac{l}{a} \times \frac{1000}{N}$$

$$100 = \frac{1}{R} \times \frac{0.5}{1.5} \times \frac{1000}{0.1}$$

$$\boxed{R = \frac{100}{3}}$$
 $I = \frac{V}{R} = \frac{5 \times 3}{100} = \frac{15}{100} = \boxed{0.15 \text{ Amp}}$

Qsn The conductivity of 0.1N NaOH solution is 0.022 scm^{-1} . When equal volume of 0.1N HCl solution is added. The conductivity of resultant solution decreases to 0.0055 scm^{-1} . Then calculate equivalent conductivity of resultant solution



0.1N 0.1N

let vol^m = 1lit 1lit

gm eq = N x V N x V
= 0.1 gmeq 0.1 gmeq

After Reaction = 0 0

$$N_{\text{NaCl}} = \frac{0.1 \text{ gmeq}}{(1+1) \text{ lit}} \\ = 0.05 \text{ N}$$

$$\Lambda_{eq} = K \times \frac{1000}{N}$$

$$\Lambda_{eq} = 0.0055 \times \frac{1000}{0.050}$$

$$\boxed{\Lambda_{eq} = 110}$$

Q.5 When a certain conductance was filled with 0.05 M KCl solution it had resistance of $400\ \Omega$ at 25°C when it is filled with CaCl_2 solution (11.1 gm in 500 mL solution). it had resistance of $900\ \Omega$ if conductivity of 0.05 N KCl solution is 0.00189 S cm^{-1} calculate (Ans = 0.756)

- (a) cell constant (Ans = 0.756)
- (b) Conductivity of CaCl_2 solution (Ans = $8.4 \times 10^{-4}\text{ S cm}^{-1}$)
- (c) molar conductivity of CaCl_2 solution (Ans = $4.2\text{ S cm}^2\text{ mol}^{-1}$)
- (d) equivalent conductivity of CaCl_2 solution (Ans = 2.1 S cm)

FACTOR EFFECTING ELECTROLYTIC CONDUCTANCE

- [I] INTER IONIC INTERACTION
- [II] SOLVENT OR HYDRATION OF IONS
- [III] VISCOSITY
- [IV] TEMPERATURE
- [V] DILUTION

[I] INTER IONIC INTERACTION

If inter ionic i. is more then mobility of ions will be decreases and thus conductance will decreases

[II] SOLVATION OR HYDRATION

* Solvent molecule का ION से जुड़ना solvation कहलाता है। अब जब ION को solvent पकड़ कर रखेगा तो ION move ही नहीं कर पाएगा या अगर move करेगा तो slowly move करेगा so conductance decreases.

* जो ion जितना छोटा होता है पानी से उतना ही ज्यादा मोटा हो जाता है पानी पीकर नहीं बल्कि H_2O को जोड़कर

* SIZE OF ION = $Li^+ < Na^+ < K^+ < Rb^+ < Cs^+$

* HYDRATED SIZE = $Li^+ > Na^+ > K^+ > Rb^+ > Cs^+$

[III] VISCOSITY

VISCOSITY बढ़ेगा तो IONS easily move नहीं कर पाएगा
so VISCOSITY OF MEDIUM INCREASES, conductance decreases

[IV] TEMPERATURE

on increasing Temp K.E of ions increases and force of attraction become weak therefore mobility of ions increases. that's why -

Temperature increases $(T) \uparrow = \text{conductance } (G) \uparrow$

(iv) DILUTION

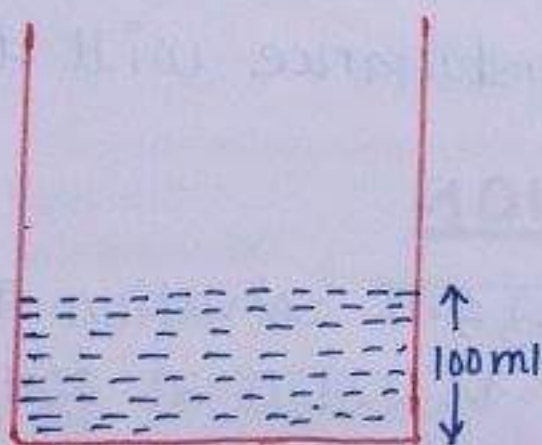
अब तक जो तीन effect था interionic interaction, solvation and temperature से तीनो common concept था. But effect of dilution on conductance is so important concept.

DILUTION INCREASES = CONDUCTANCE INCREASES (G)

DILUTION INCREASES = CONDUCTIVITY DECREASES (K)

BEFORE DILUTION

1 mol present in 100 ml



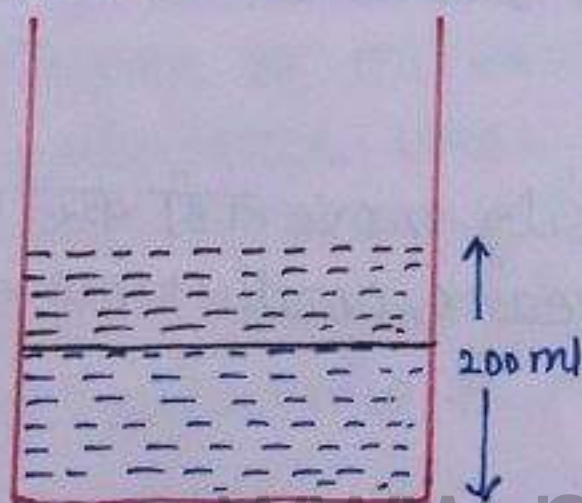
Let

100 ml solⁿ have = 50,00,000 ions = $G_1 = \Lambda M_1$

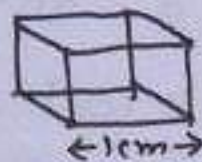
1 ml solⁿ have = 50,00,000 ions = K_1 [K Always 1ml के लिए होता है]

AFTER DILUTION

1 mol in 200 ml



1 ml =



$G_2 > G_1$ $K_1 > K_2$

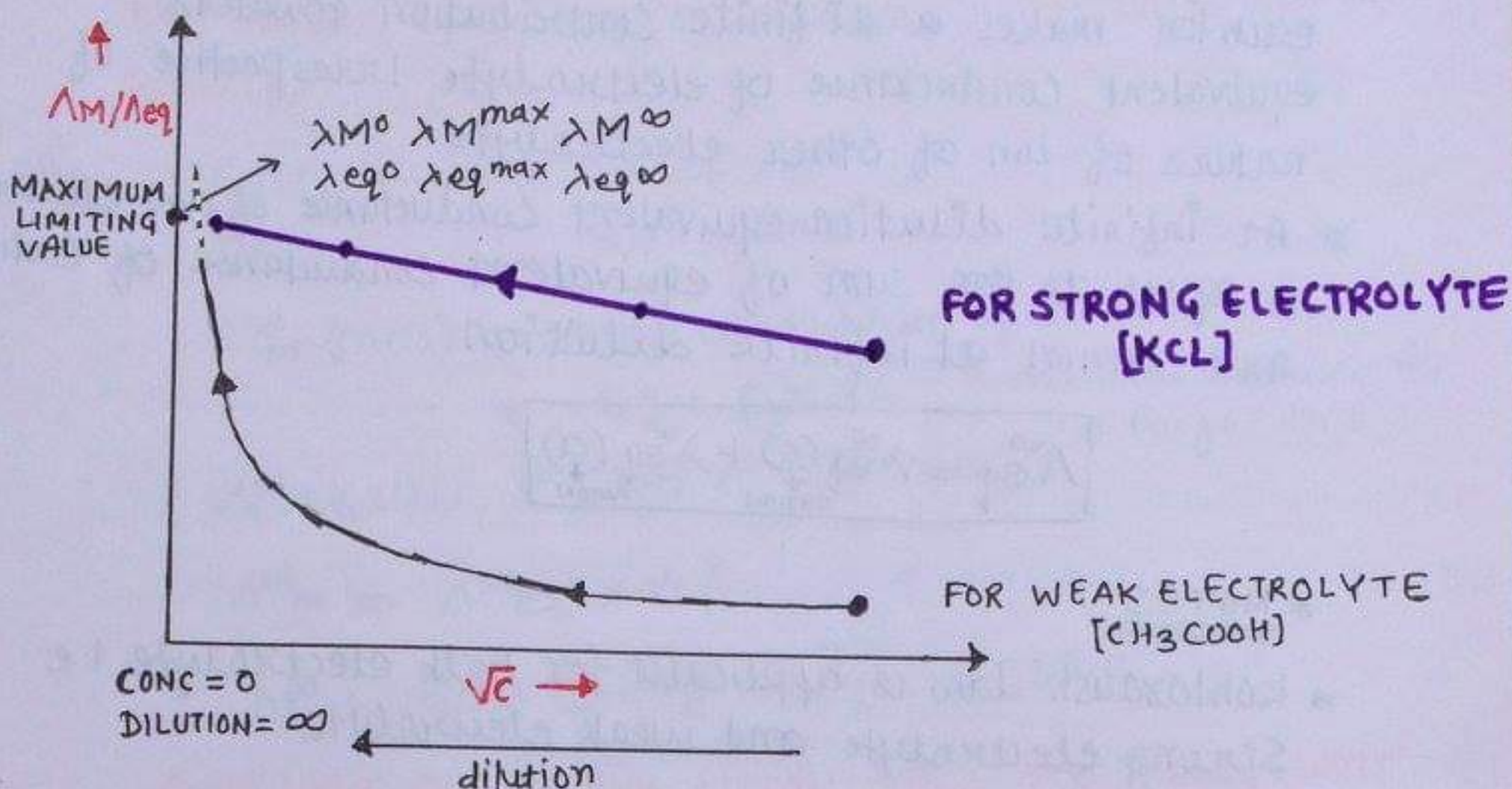
200 ml solⁿ contains 60,00,000 ions = $G_2 = \Lambda M_2$

1 ml solⁿ contains 30,00,000 ions = K_2

* ON DILUTION $\Rightarrow G \uparrow$ BUT $K \downarrow$

* ON DILUTION $\Rightarrow \Lambda_{eq} \uparrow$ AND $\Lambda_M \uparrow$ [दोनों INCREASES]

DEBYE-HUCKEL-ONSAGAR GRAPH



FOR STRONG ELECTROLYTE

$$\Lambda_m = \Lambda_m^\infty - b\sqrt{c}$$

OR

$$\Lambda_{eq} = \Lambda_{eq}^\infty - b\sqrt{c}$$

→ DEBYE-HUCKEL-ONSAGAR EQUATION

→ b = DHO constant

→ b depends upon - Nature of electrolyte
 - Nature of solvent
 - Temperature

OBSERVATION OF DHO-GRAPH

- * Dilution ko increase करने पर strong electrolyte की molar conductance weak electrolyte की molar conductance के जितना increase नहीं होता.
 BUT strong electrolyte की conductance at weak electrolyte की conductance से Always greater ही होगा.
- * on increase dilution conduction of strong electrolyte तो gradually increase होती है। But conduction of weak electrolyte rapidly increase करती है।
- * अगर Graph को extrapolate किया जाए तो strong electrolyte की λ_m obtain किया जा सकता है। But weak electrolyte की maximum λ_m obtain नहीं किया जा सकता है, क्योंकि weak electrolyte की α is always less than 1

KOHLRAUSCH'S LAW

- * At infinite dilution when dissociation is completed each ion makes a definite contribution towards equivalent conductance of electrolyte irrespective of nature of ion of other electrolyte
- * At infinite dilution equivalent conductance of an electrolyte is equal to the sum of equivalent conductance of cation and anion at infinite dilution

$$\Lambda_{eq}^{\infty} = \lambda_{eq}^{\infty}(\underset{\text{CATION}}{c}) + \lambda_{eq}^{\infty}(\underset{\text{ANION}}{a})$$

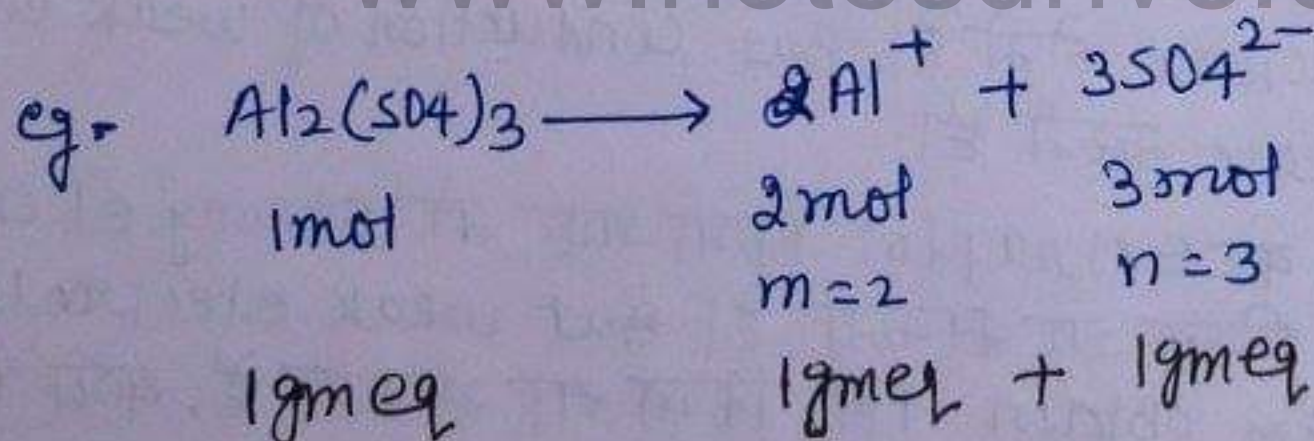
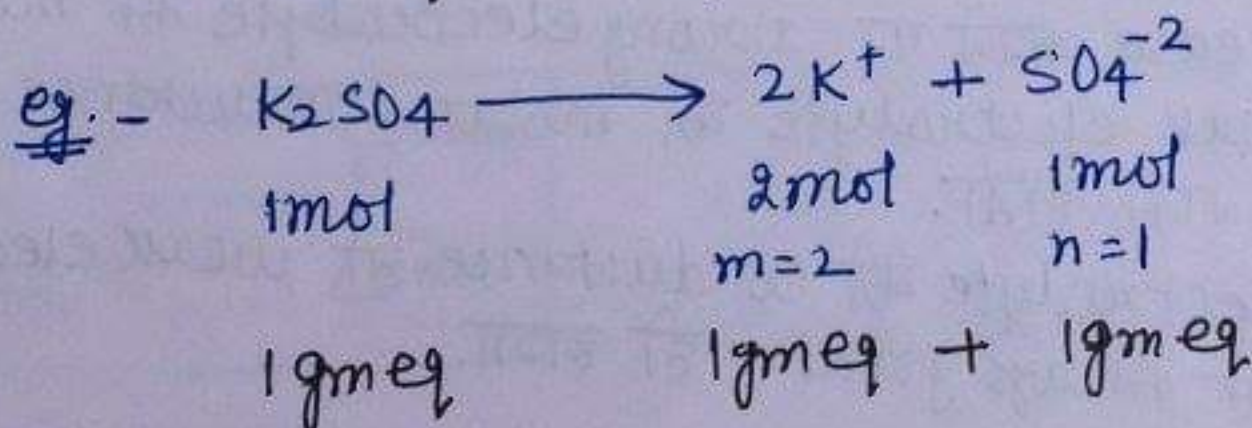
* NOTES

- ⇒ Kohlrausch law is applicable for both electrolyte i.e. strong electrolyte and weak electrolyte
- ⇒ Kohlrausch's original statement was in terms of equivalent conductance but according to I.U.P.A.C. convention now Kohlrausch law is expressed in terms of molar conductance

$$\Lambda_M^{\infty} = m \lambda_M^{\infty}(c) + n \lambda_M^{\infty}(a)$$

m = No of moles of cation in 1 mol of electrolyte

n = No of moles of anion in 1 mol of electrolyte



Qsn ionic conductance at infinite dilution of Ca^{2+} and Cl^- are 119 and 76 $\text{cm}^2 \text{mol}^{-1}$ respectively. Calculate the molar equivalent conductance of CaCl_2 at infinite dilution

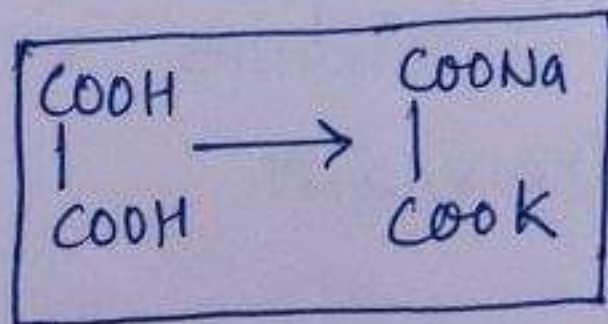
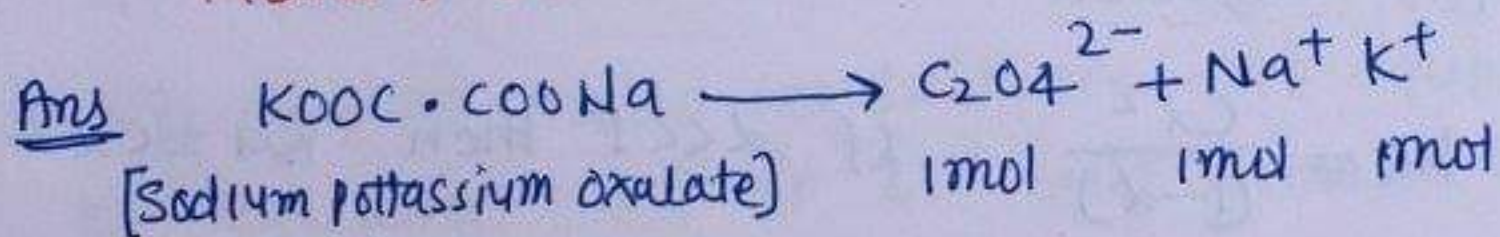


$\Lambda_M^\infty(\text{CaCl}_2) = \text{molar conductance of cation} + \text{Anion}$
 $= 119 + 2 \times 76$ → maximum mistake we forget this
 $\Lambda_M^\infty(\text{CaCl}_2) = 271 \text{ } \Omega^{-1} \text{cm}^2 \text{mol}^{-1}$

$\Lambda_M^\infty = \Lambda_{eq}^\infty \times v.f$

$\Lambda_{eq}^\infty(\text{CaCl}_2) = \frac{\Lambda_M^\infty}{v.f} = \frac{271}{2} = 135.5$

Qsn Calculate the molar conductance and equivalent conductance at infinite dilution of the salt $\text{KOOx} \cdot \text{COONa}$. Given the ionic conductance is at infinite dilution O^{2-} (oxalate ion), K^+ and Na^+ are 148.2, 50.1 and 73.5 $\Omega^{-1} \text{cm}^2 \text{mol}^{-1}$



$\Lambda_M^\infty = \Lambda_M^\infty(\text{Ox}^{2-}) + \Lambda_M^\infty \text{Na}^+ + \Lambda_M^\infty \text{K}^+$

$\Lambda_M^\infty = 148.2 + 50.1 + 73.5$

$\Lambda_M^\infty = 271.8 \text{ } \Omega^{-1} \text{cm}^2 \text{mol}^{-1}$

$\Lambda_{eq}^\infty = \frac{\Lambda_M^\infty}{v.f} = \frac{271.8}{2} = 135.9$

APPLICATION OF KOHLRAUSH LAW

- [I] To Determine degree of ionisation (α)
- [II] To determine dissociation constant [K]
- [III] To determine Λ_m^∞ OR λ_{eq}^∞ for weak electrolyte
- [iv] To determine solubility of sparingly soluble salt

[I] TO DETERMINE DEGREE OF IONISATION

$$\alpha = \frac{\text{Dissociated moles}}{\text{Total moles}} = \frac{\text{ions obtained by dissociation of electrolyte}}{\text{Total ions present in electrolyte}}$$

At constant "concentration c"

$$\alpha = \frac{\Lambda_{eq} c}{\Lambda_{eq}^\infty} \text{ or } \frac{\Lambda_m c}{\Lambda_m^\infty}$$

[II] TO DETERMINE DISSOCIATION CONSTANT [K]

eg - for CH_3COOH

$$K_a = \frac{c\alpha^2}{(1-\alpha)} \quad \text{if } \alpha \ll 1 \text{ then } K_a = c\alpha^2$$

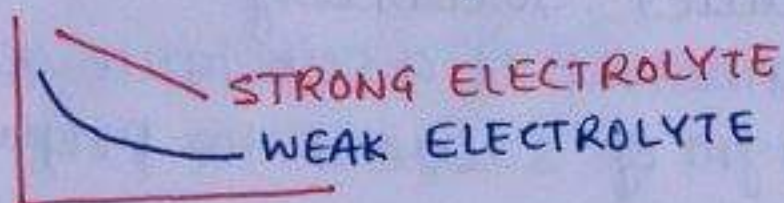
$$[\text{H}^+] = c\alpha$$

$$i = 1 - \alpha + n\alpha$$

[III] TO DETERMINE Λ_M^∞ OR Λ_{eq}^∞ FOR A WEAK ELECTROLYTE

Q.57: $\Lambda_M^\infty (\text{CH}_3\text{COOH}) = ?$

- * यहाँ CH_3COOH एक weak electrolyte है। और graph से weak electrolyte का Λ_M^∞ find नहीं कर सकते हैं।
Graph से तो सिर्फ strong electrolyte का ही Λ_M^∞ find कर सकते



- * But we can use Kohlrausch law because Kohlrausch law is applicable for both strong and weak electrolyte

$$\Lambda_M^\infty (\text{CH}_3\text{COOH}) = \Lambda_M^\infty (\text{CH}_3\text{COO}^-) + \Lambda_M^\infty (\text{H}^+) \quad \text{--- (i)}$$

$$\Lambda_M^\infty (\text{CH}_3\text{COONa}) = \Lambda_M^\infty (\text{CH}_3\text{COO}^-) + \Lambda_M^\infty (\text{Na}^+) \quad \text{--- (ii)}$$

$$\Lambda_M^\infty (\text{HCl}) = \Lambda_M^\infty (\text{H}^+) + \Lambda_M^\infty (\text{Cl}^-) \quad \text{--- (iii)}$$

$$\Lambda_M^\infty (\text{NaCl}) = \Lambda_M^\infty (\text{Na}^+) + \Lambda_M^\infty (\text{Cl}^-) \quad \text{--- (iv)}$$

- * $\Lambda_M^\infty (\text{CH}_3\text{COO}^-)$ की molar conductance चाहे equation (i) में हो या equation (ii) में हो दोनों जगह same ही होगी

- * $\Lambda_M^\infty (\text{H}^+)$ की molar conductance चाहे equation (i) में हो या equation (iii) में हो दोनों जगह same ही होगी

- * equation (i) = equation (ii) + equation (iii) - equation (iv)

$$\Lambda_M^\infty (\text{CH}_3\text{COOH}) = \Lambda_M^\infty (\text{CH}_3\text{COO}^-) + \Lambda_M^\infty (\text{H}^+) - \Lambda_M^\infty (\text{NaCl})$$

$$\Lambda_M^\infty (\text{CH}_3\text{COOH}) = \Lambda_M^\infty (\text{CH}_3\text{COONa}) + \Lambda_M^\infty (\text{HCl}) - \Lambda_M^\infty (\text{NaCl})$$

[IV] TO DETERMINE SOLUBILITY OF SPARINGLY SOLUBLE SALT

Eg - AgCl , BaSO_4 , PbSO_4

Solubility

- Concentration of saturated solution at constant temperature called solubility
- The maximum amount of a substance which can be dissolve in 100 gm of solvent to prepare its saturated solution

FUNDA:

- saturated solution of sparingly soluble salt can be considered as infinitely diluted solution because these salt dissolve very less therefore

Saturated solution = infinitely diluted solution

$$\Lambda_M = \Lambda_M^\infty = k \times \frac{1000}{M}$$

↓
By Kohlrausch law

यहाँ पर $M=S$

* if we add 500 gm AgCl in 1 litre of H_2O then only upto 5g AgCl is dissolve.

* 495 gm AgCl remain undissolve. so actual solubility = 5g/l

Qsn At a certain temperature the saturated solution of AgCl has conductivity (κ) $1.8 \times 10^{-6} \text{ mho cm}^{-1}$. The ionic conductance of Ag^+ and Cl^- at infinite dilution are 54.5 and 65.5 $\text{mho cm}^2 \text{ mol}^{-1}$. Find the solubility of AgCl at this temperature

Ans $\Lambda_M^\infty \text{ AgCl} = 54.5 + 65.5$
 $= 120$

$$\Lambda_M^\infty = 1.8 \times 10^{-6} \times \frac{1000}{S}$$

$$120 = 1.8 \times 10^{-6} \times \frac{1000}{S}$$

$$S = \frac{1.8 \times 10^{-6} \times 1000}{120}$$

$$S = 1.5 \times 10^{-5} \text{ mol/litre}$$

$$K_{sp} = S^2 = 2.25 \times 10^{-10}$$

Qsn The molar conductance for acetic acid at 0.2 M molar conductance at infinite dilute are 4.8 and $400 \text{ S}^{-1} \text{ cm}^2 \text{ mol}^{-1}$ respectively. Calculate degree of dissociation and dissociation constant of CH_3COOH .

Ans $C = 0.2 \text{ M } \text{CH}_3\text{COOH}$

$$\Lambda_M^c = 4.8$$

$$\Lambda_M^\infty = 400 \text{ S}^{-1} \text{ cm}^2 \text{ mol}^{-1}$$

$$\alpha = \frac{\Lambda_M^c}{\Lambda_M^\infty} = \frac{4.8}{400} \times 100 = 1.2\% = 1.2 \times 10^{-2}$$

$$K_a = C\alpha^2$$

$$= 0.2 (1.2 \times 10^{-2})^2$$

$$= 2.88 \times 10^{-5}$$

Qsn The molar conductance of an infinitely diluted solution of Ammonium chloride (NH_4Cl) is $150 \Omega^{-1} \text{cm}^2 \text{mol}^{-1}$ and ionic conductance is ($\text{OH}^- = 198$), ($\text{Cl}^- = 76 \Omega^{-1} \text{cm}^2 \text{mol}^{-1}$) What will be the molar conductance of the solution of NH_4OH (Ammonium hydroxide) at infinite dilution. If the conductivity of 0.01M solution of NH_4OH is $9.52 \times 10^{-5} \Omega^{-1} \text{cm}^2$ what will be its degree of dissociation.

Ans $\text{NH}_4\text{Cl} = 150 \Omega^{-1} \text{cm}^2 \text{mol}^{-1}$

$\text{OH}^- = 198$

$\text{Cl}^- = 76 \Omega^{-1} \text{cm}^2 \text{mol}^{-1}$

$\Lambda^\infty \text{M} (\text{NH}_4\text{OH}) = \Lambda^\infty \text{M} (\text{NH}_4\text{Cl} + \text{OH}^- - \text{Cl}^-)$

$\Lambda^\infty \text{M} (\text{NH}_4\text{OH}) = \Lambda^\infty \text{M} (150 + 198 - 76)$

$\Lambda^\infty \text{M} (\text{NH}_4\text{OH}) = 272 \Omega^{-1} \text{cm}^2 \text{mol}^{-1}$

$C = 0.01 \text{M}$

$K = 9.52 \times 10^{-5} \Omega^{-1} \text{cm}^2$

$\Lambda^c \text{M} = K \times \frac{1000}{M}$

$\Lambda^c \text{M} = \frac{9.52 \times 10^{-5} \times 1000}{0.01} = 9.52$

$\alpha = \frac{\Lambda^c \text{M}}{\Lambda^\infty \text{M}} = \frac{9.52}{272} \times 100$

$\alpha = 3.5\%$

Qsn Conductivity of saturated solution of AgCl is $2.2 \times 10^{-6} \Omega^{-1} \text{cm}^{-1}$ and conductivity of water is $0.8 \times 10^{-6} \Omega^{-1} \text{cm}^{-1}$. Molar conductance at infinite dilution of AgNO₃, HCl and HNO₃ are 135, 426 and 421 $\Omega^{-1} \text{cm}^2 \text{mol}^{-1}$ respectively calculate solubility product of AgCl

Ans

$$\begin{aligned}\text{AgNO}_3 &= 135 \\ \text{HCl} &= 426 \\ \text{HNO}_3 &= 421\end{aligned}$$

$$\begin{aligned}\Lambda_m^\infty(\text{AgCl}) &= \Lambda_m^\infty(\text{AgNO}_3 + \text{HCl} - \text{HNO}_3) \\ &= \Lambda_m^\infty(135 + 426 - 421)\end{aligned}$$

$$\Lambda_m^\infty(\text{AgCl}) = 140 \Omega^{-1} \text{cm}^2 \text{mol}^{-1}$$

$$K_{\text{AgCl}} \text{ solution} = 2.2 \times 10^{-6} \Omega^{-1} \text{cm}^{-1}$$

$$K_{\text{H}_2\text{O}} \text{ solution} = 0.8 \times 10^{-6} \Omega^{-1} \text{cm}^{-1}$$

$$K_{\text{solution of electrolyte}} = K_{\text{electrolyte}} + K_{\text{H}_2\text{O}}$$

$$= K_{\text{solute electrolyte}} + K_{\text{H}_2\text{O}}$$

$$= 2.2 \times 10^{-6} - 0.8 \times 10^{-6}$$

$$K_{\text{AgCl}} = 1.4 \times 10^{-6}$$

it should neglect

Qsn specific conductance of 0.03M aqueous solution of weak base BOH is $1.5 \times 10^{-4} \text{Scm}^{-1}$. also the molar conductance at infinite dilution is $250 \text{Scm}^2 \text{mol}^{-1}$ find ionisation constant of weak base

Ans

$$\begin{aligned}C &= 0.03 \text{M} \\ K &= 1.5 \times 10^{-4} \text{Scm}^{-1} \\ \Lambda_m^\infty &= 250 \text{Scm}^2 \text{mol}^{-1}\end{aligned}$$

$$\alpha = \frac{\Lambda_m C}{\Lambda_m^\infty} = \frac{5}{250} = 0.02$$

$$K_b = \frac{C\alpha^2}{1-\alpha} = C\alpha^2$$

$$K_b = 0.03 (0.02)^2$$

$$K_b = 1.2 \times 10^{-5}$$

$$\Lambda_m^c = K \times \frac{1000}{M}$$

$$\Lambda_m^c = 1.5 \times 10^{-4} \times \frac{1000}{0.03}$$

$$\Lambda_m^c = 5$$

AIMS

ELECTROLYTE	KCl	KNO ₃	HCl	NaOAc	NaCl
Λ^∞ ($\text{Scm}^2\text{mol}^{-1}$)	149.9	145.0	426.2	91.0	126.5

calculate $\Lambda^\infty_{\text{HOAc}}$ using appropriate molar conductance of electrolyte listed above at infinite dilution in H_2O at 25°C

Ans

$$\begin{aligned}\Lambda^\infty_{\text{HOAc}} &= \Lambda^\infty_{\text{NaOAc}} + \Lambda^\infty_{\text{HCl}} - \Lambda^\infty_{\text{NaCl}} \\ &= 91.0 + 426.2 - 126.5 \text{ Scm}^2\text{mol}^{-1} \\ &= 390.7 \text{ Scm}^2\text{mol}^{-1}\end{aligned}$$

Qsn
AIMS

Hydrofluoric acid is weak acid. At 25°C the molar conductivity of 0.002 M HF is $176.2 \text{ S}^{-1}\text{cm}^2\text{mol}^{-1}$. Equilibrium if its $\Lambda^\infty_m = 405.2 \text{ S}^{-1}\text{cm}^2\text{mol}^{-1}$, Equilibrium constant at the given concentration is

Ans

$$\alpha = \frac{\Lambda_c}{\Lambda^\infty_m} = \frac{176.2 \text{ S}^{-1}\text{cm}^2\text{mol}^{-1}}{405.2 \text{ S}^{-1}\text{cm}^2\text{mol}^{-1}} = 0.435$$

$$K = \frac{[\text{H}^+][\text{F}^-]}{[\text{HF}]} = \frac{c\alpha^2}{1-\alpha} = \frac{(0.002 \text{ M})(0.435)^2}{1-0.435}$$

$$K = 6.70 \times 10^{-4} \text{ M}$$

ELECTROCHEMISTRY - PART TWO

ELECTRODE POTENTIAL

When a metal is placed in aqueous solution of its ion it acquires either negative or positive charge with respect to solution and potential is developed between metal and solution which is called electrode potential of metal.

Explanation:

- * अगर Cu को उसके aqueous solution CuSO_4 में डुबाया जाए तो potential developed हो जाएगा.
- * अगर Zn को उसके aqueous solution ZnSO_4 में डुबाया जाए तो potential developed हो जाएगा.

Electrode potential mainly depends upon

- (a) Nature of electrode
- (b) Concentration of ions in solution
- (c) Temperature

NCERT

A large number of

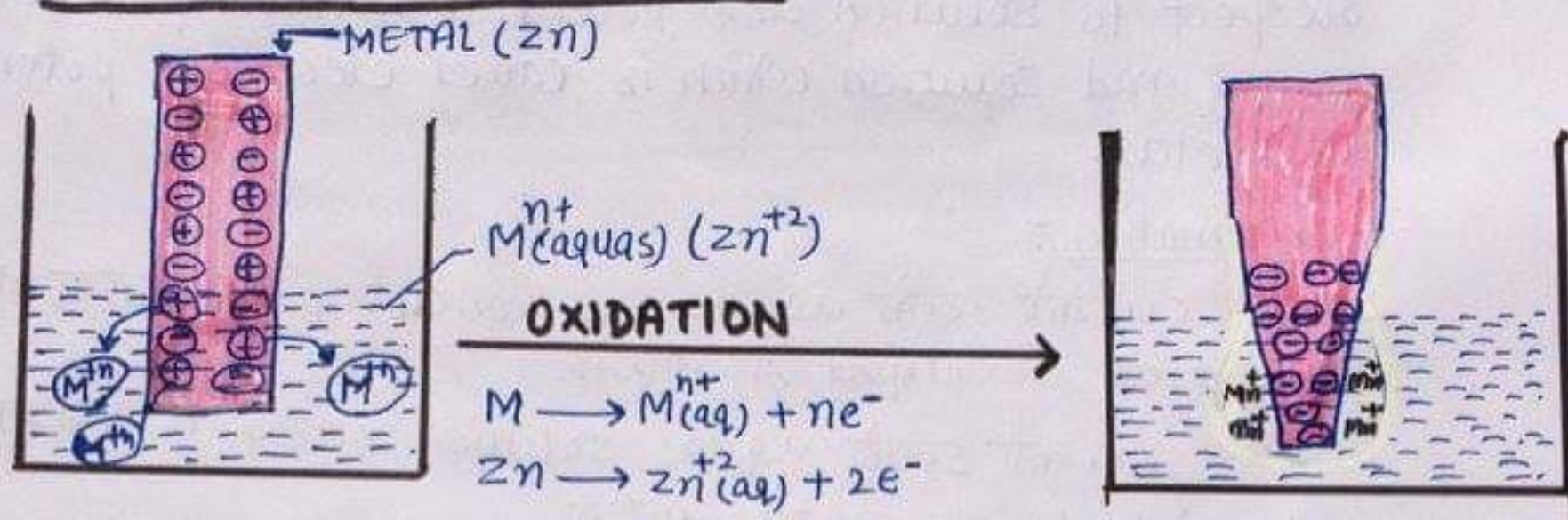
{	Metals
	Sodium hydroxide
	Fluorine
	Chlorine
	and many other chemicals

 are produced by electrochemical method.

ELECTRODE POTENTIAL ARE OF TWO TYPES:

- (i) OXIDATION POTENTIAL
- (ii) REDUCTION POTENTIAL

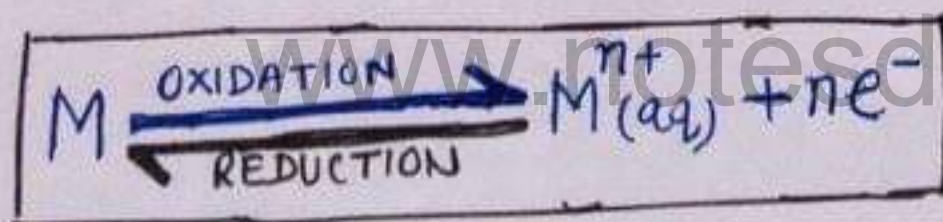
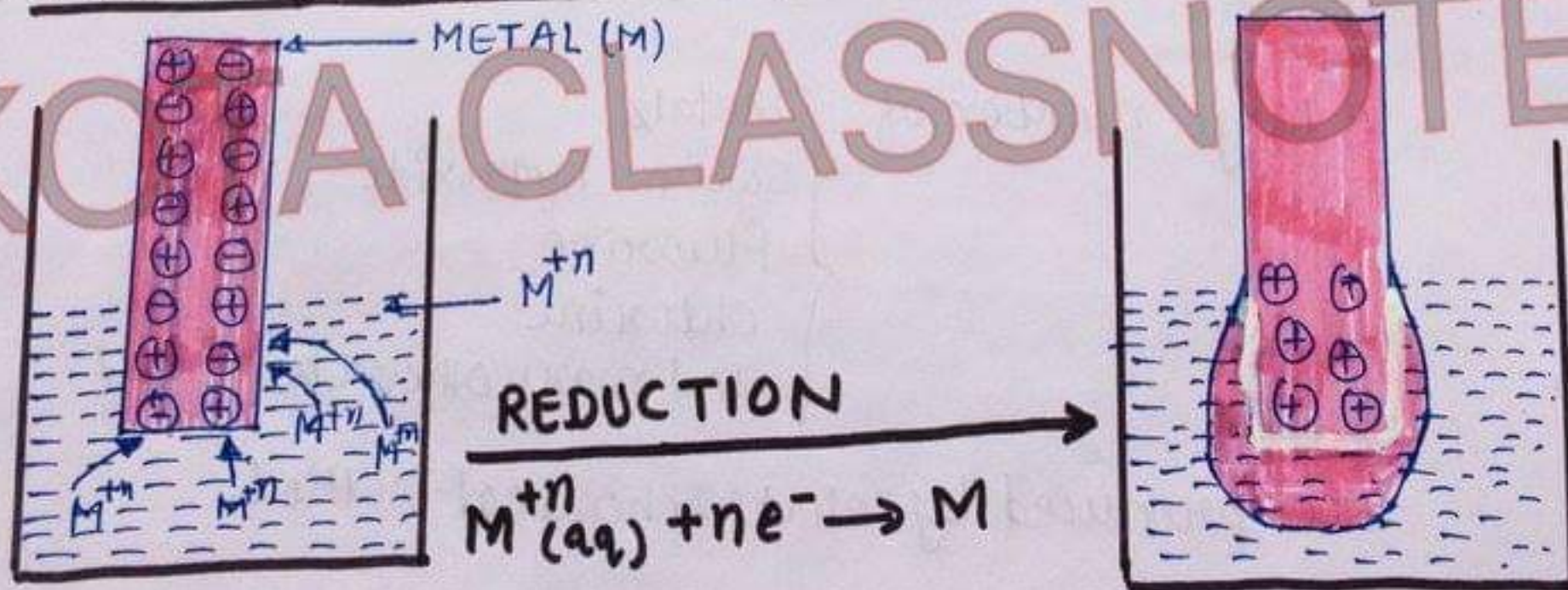
(i) OXIDATION POTENTIAL



* Metal atom of electrode have tendency to go into the solution as ions. and leave behind the electrons at the electrode trying make it negatively charged.

* यहाँ electrode का oxidation हो रहा है। और potential develop हो रहा है। so called oxidation potential

(ii) REDUCTION POTENTIAL



IF OXID > RED THEN OXID-POTENTIAL
IF RED > OXID THEN RED-POTENTIAL

REFERENCE ELECTRODE

कोई भी Battery at least दो Half cell की बनी होती है।

But when we want measure potential of single half cell हम measure नहीं कर पाएंगे इसके लिए reference electrode की requirement होगी

PRIMARY REFERENCE ELECTRODE [SHE/NHE]

SHE = Standard Hydrogen electrode

Hydrogen electrode represented by = $\text{Pt(s)} | \text{H}_2(\text{g}) | \text{H}^+(\text{aq})$

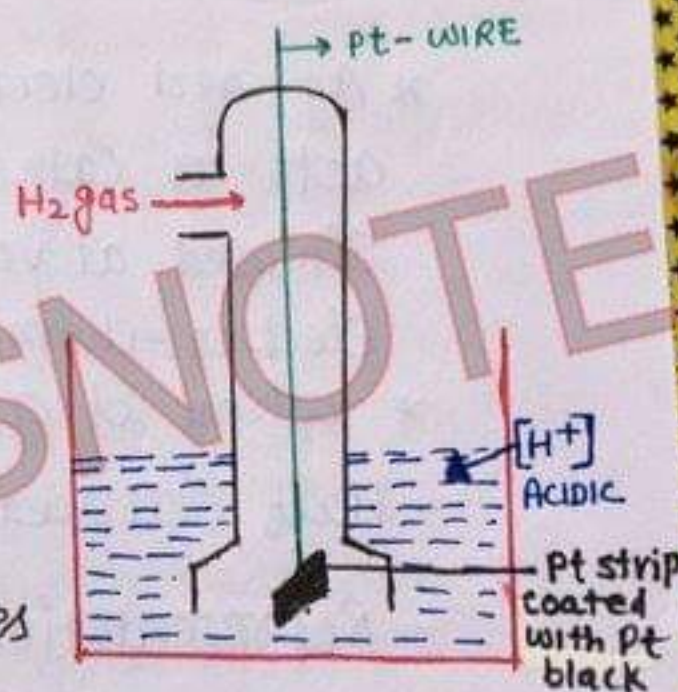
It is assigned to ZERO POTENTIAL at all temp corresponding to this reaction



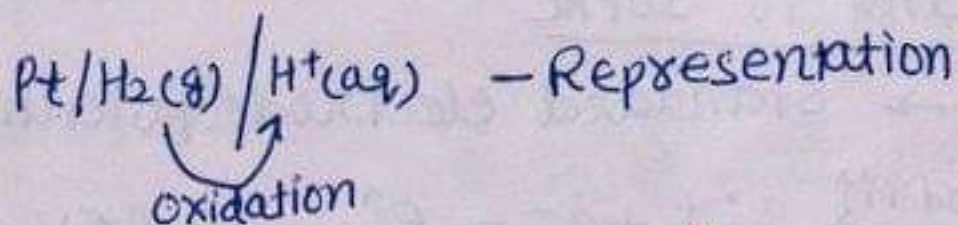
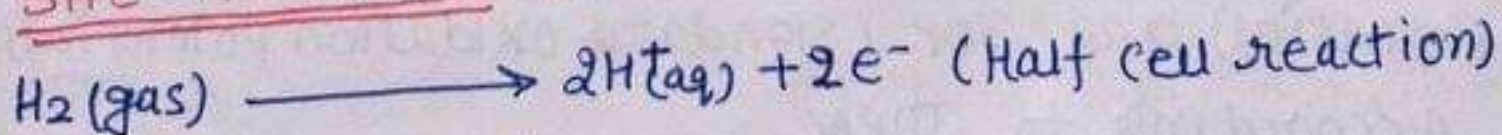
* H_2 is a gas so gas wants free and to hold gas Pt-strip is coated with Pt-black (Powder)

* Pt is black powder which increases surface area.

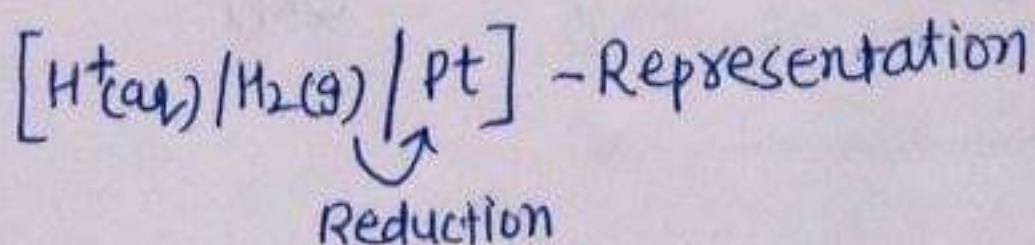
* Pt is less reactive and cannot react with H^+ (Acid)



SHE AS ANODE (OXIDATION)

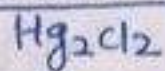


SHE AS CATHODE (REDUCTION)



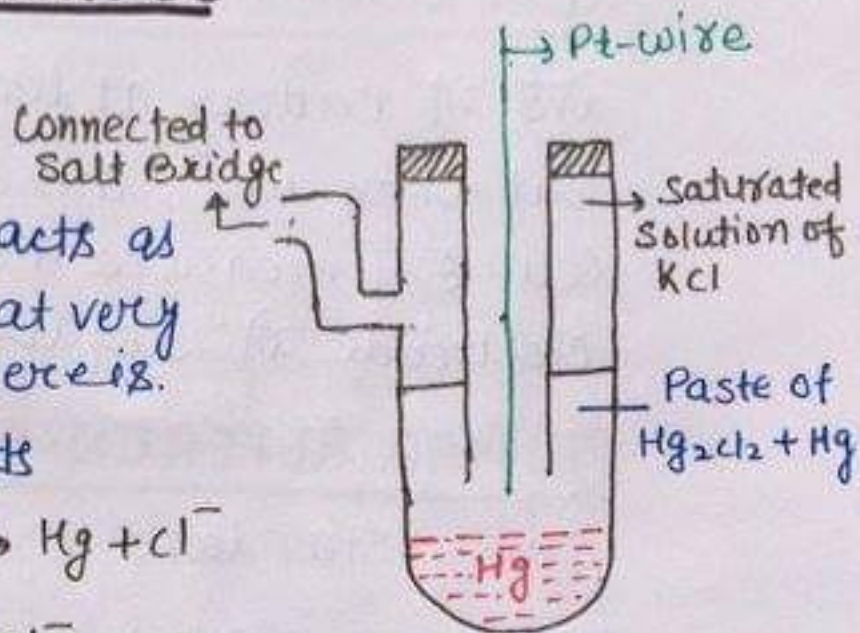
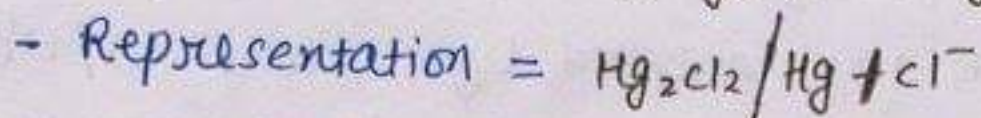
SECONDARY REFERENCE ELECTRODE

[A] CALOMEL ELECTRODE



* Calomel electrode is commonly acts as Cathode because it is situated at very low position in Electrochemical series.

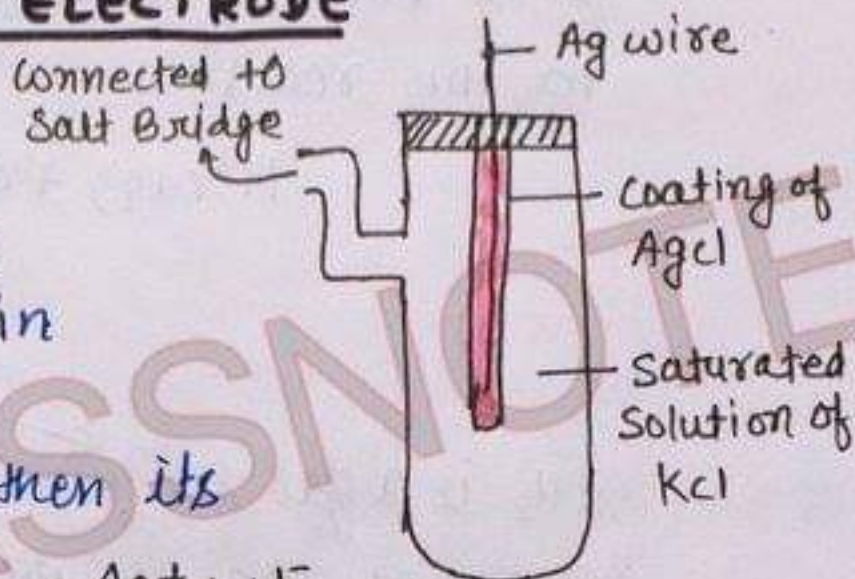
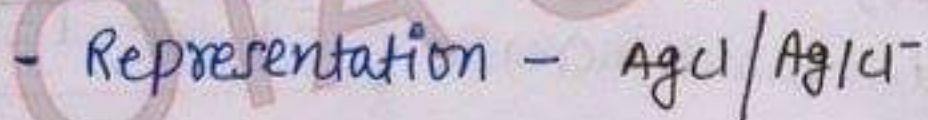
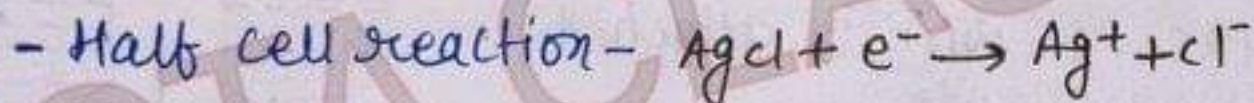
* When it acts as cathode then its



[B] SILVER-SILVER CHLORIDE ELECTRODE

* Ag-AgCl electrode generally acts as Cathode because it is situated at very low position in Electro-chemical series

* When it acts as cathode then its

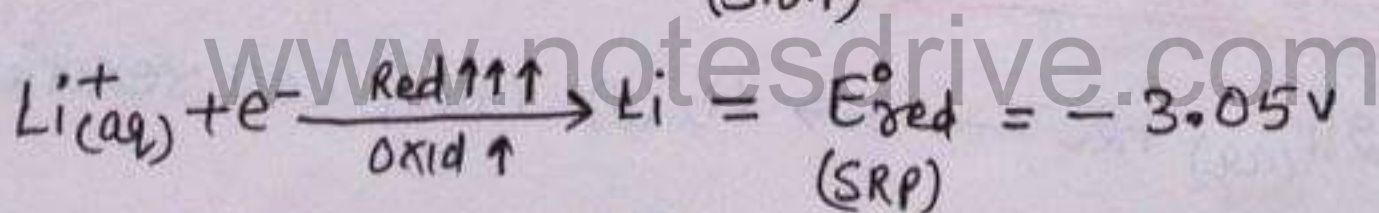
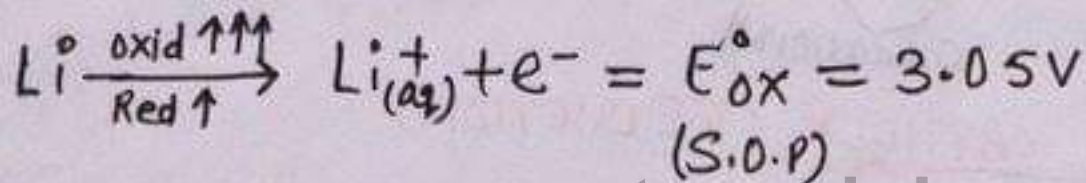


NOTE

⇒ For secondary reference electrode SRP (standard reduction potential) and SOP (standard oxidation potential) not zero

⇒ According to IUPAC

SRP $\longleftrightarrow$ standard electrode potential (E°)



ELECTROCHEMICAL SERIES

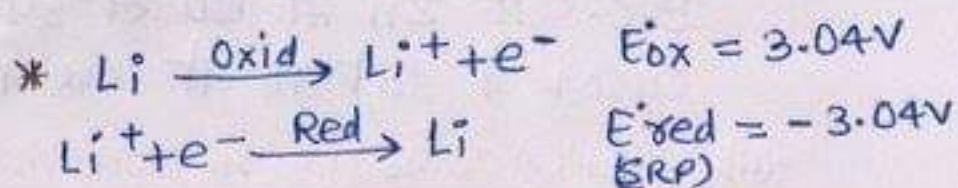
लीथम Li = SRP = -3.04V Best reducing agent

के	K
बाप	Ba
से	Sr
काका	Ca
ने	Na
माँगी	Mg
अलमारी	Al
मनी	M
जमीन	Zn = -0.76V
कार	Cu
फ्रिज	Fe
कड़े	Cd
को	Co
नी	Ni
स्नेह	Sn
प्रभा	Pb
ही	H ₂ — 0 volt
कुटेगी	Cu — I ₂
होगी	Hg
आगे	Ag — Br ₂
पड़े	Pd — Cl ₂
पिलाई	Pt
आउ	Au = +1.5V

Best
Oxidising
agent

F₂ = +2.87

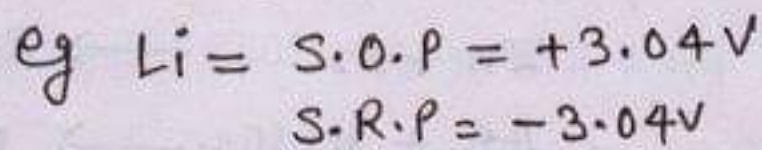
* As we go down the group
Reactivity of Non-metal increases
Reactivity of metal decreases



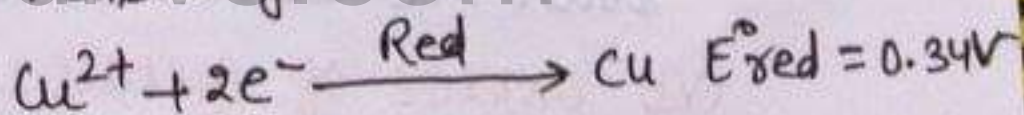
* As we go from Li to F
Strength of oxidising agent increases

* H₂ से पहले वाली को जब SHE के साथ connect करते हैं तो :-
Oxidation > Reduction
That's why oxidation potential लिखा जाएगा.

* But जब S.R.P लिखा जाएगा तो -ve sign लगेगा



* H₂ से नीचे वाली को जब SHE के साथ connect करते हैं तो :-
Reduction > Oxidation

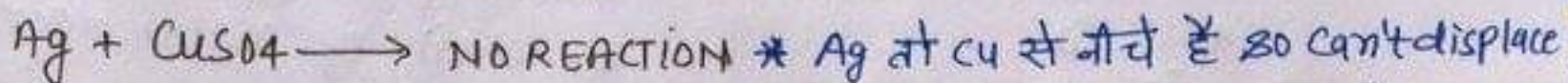


IMPORTANT CONCLUSION FROM E.C.S

(I) DISPLACEMENT OF METAL IONS BY METAL

Metal placed higher in E.C.S displaces metal ion (cation)

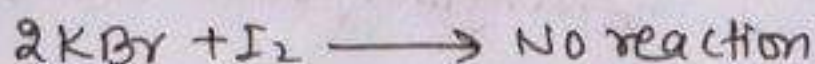
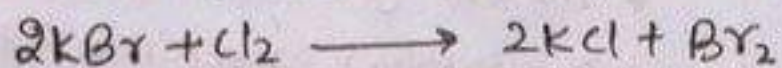
Placed below them in E.C.S from their solution



E.C.S में Zn तो Cu से ऊपर है इसलिए Zn को जब Cu के solution CuSO_4 में डालें तो वो Cu को displace कर दिगा

(II) DISPLACEMENT OF NON-METAL ION BY NON-METAL

Non-metal displaces the anion of non-metal placed above it in electrochemical series from their solution



(III) DISPLACEMENT OF H_2 GAS FROM H_2O BY METAL

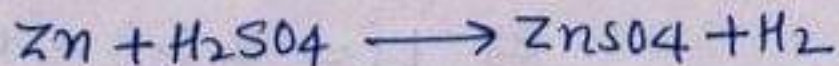
* Metal placed above Mg in E.C.S can displace H_2 gas from cold water

* Metal from Mg to Cr can displace H_2 gas from hot water

* Iron can displace H_2 gas from steam

(IV) DISPLACEMENT OF H^+ ION FROM ACIDIC SOLUTION OF METAL

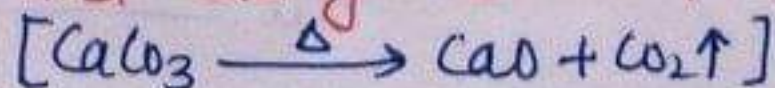
Metal placed above Hydrogen in E.C.S displaces H^+ ion from acidic solution



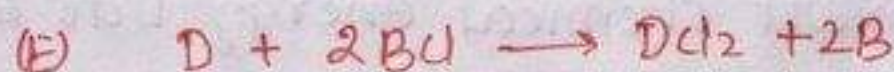
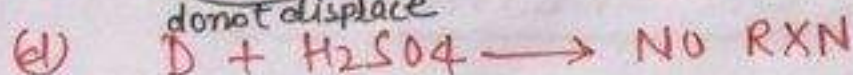
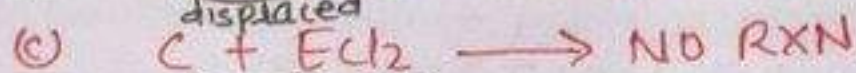
(V) STABILITY OF METAL-OXIDE DECREASES DOWN THE SERIES

If electropositive character of metal is more then the stability of metal oxide will be more so we can say that down the series stability of metal oxide decreases

(VI) oxides of metals which are placed below (Cu) are unstable and easily reduces to metal when heated



Qsn On the basis of following chemical reaction decide position of Metal ABCDE and H_2 in E.C.S



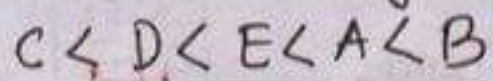
E
C
A
 H_2
D
B

SRP ↑

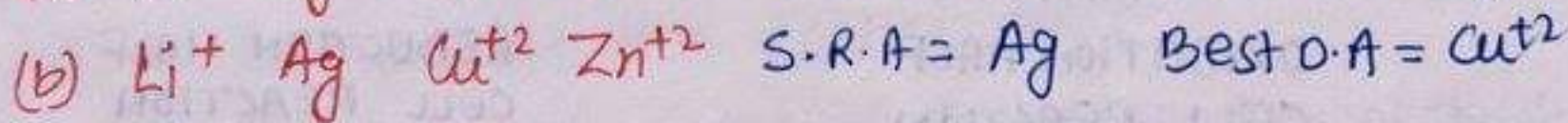
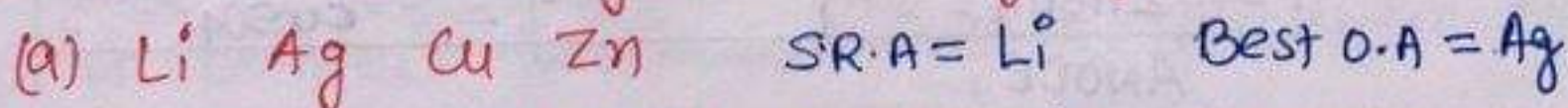
Qsn Arrange these metals in increasing order of their reducing power

METAL	E° (SRP/SEP)
A	-1.5V
B	-2.3V
C	+1.5V
D	+0.1V
E	-0.1V

* Lowest value of SRP will having highest Reducing power and weakest oxidising agent



Qsn w.o.f is strongest oxidising agent and reducing agent



ELECTROCHEMICAL CELL

There are two types of electrochemical cell

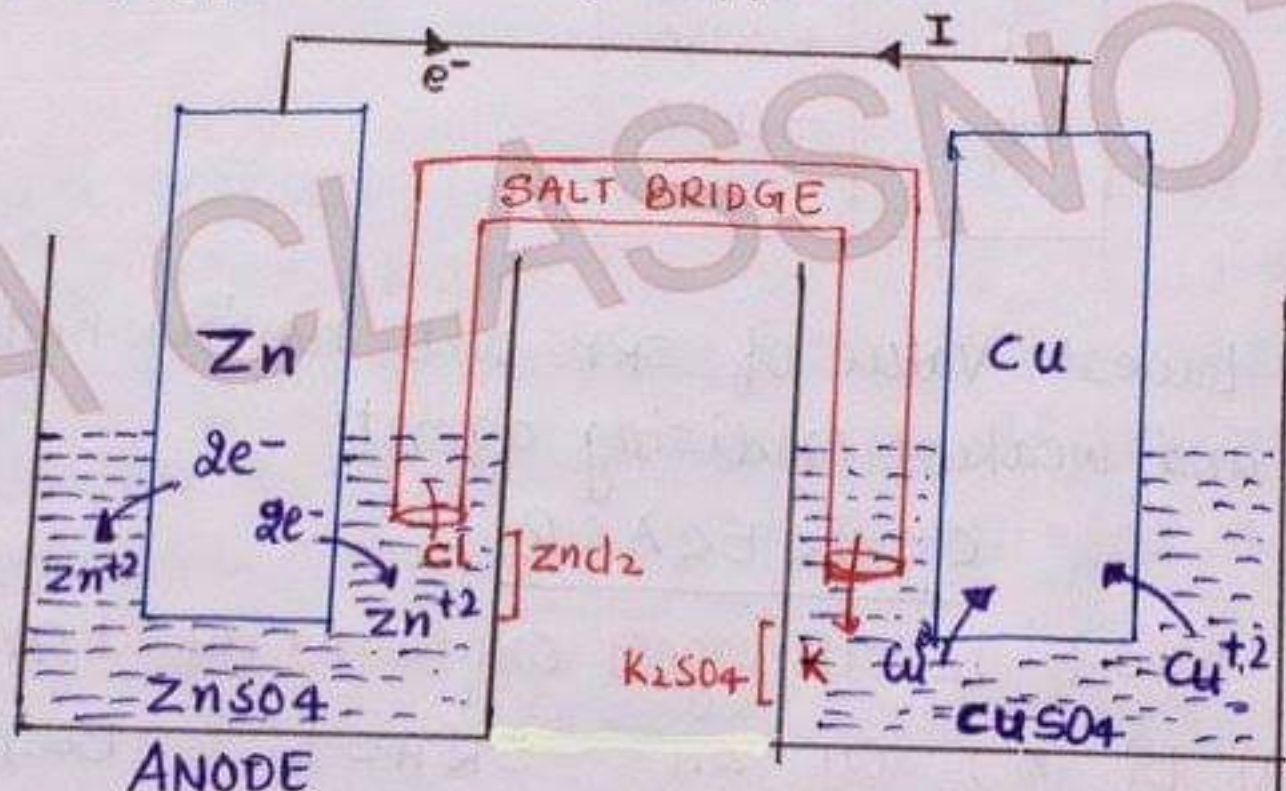
- (i) Galvanic cell OR VOLTALIC CELL
- (ii) ELECTROLYTIC CELL

(i) GALVANIC CELL OR VOLTALIC CELL

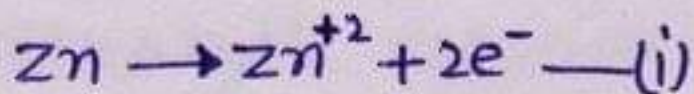
* A device which convert chemical energy into electrical energy is called galvanic cell.

* अगर Galvanic cell में Anode को Zn electrode से बनाया जाए और Cathode को Copper से बनाया जाए तब Galvanic cell DANIEL CELL कहलाएगा.

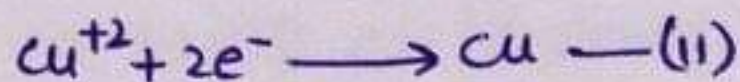
* Daniel ने जो cell बनाया था उसमें Daniel ने Cu को Cathode बनाया और dipped किया था $\text{CuSO}_4(\text{aq})$ solⁿ में Zn को anode बनाया और dipped किया था $\text{ZnSO}_4(\text{aq})$ में



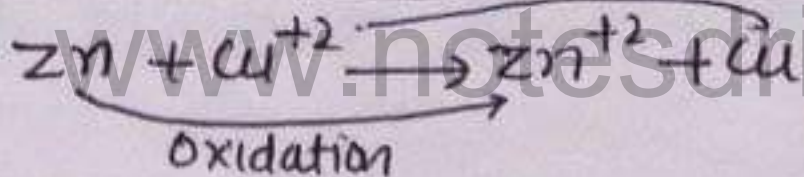
ANODE
OXIDATION HALF
CELL REACTION



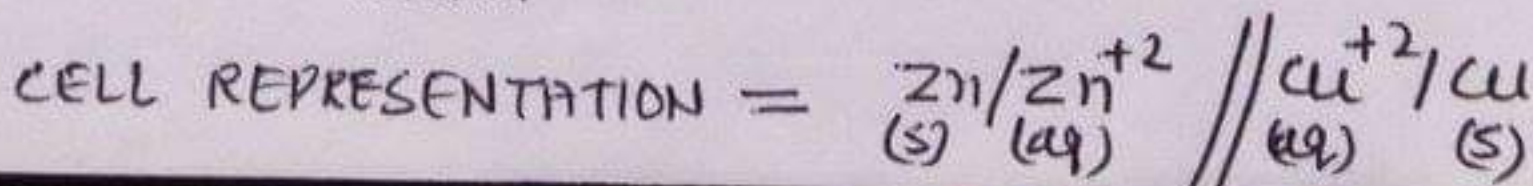
CATHODE
REDUCTION HALF
CELL REACTION



COMPLETE CELL REACTION



Reduction because Cu की S.R.P. ज्यादा



Following observations are observed during cell reaction

- (i) During cell reaction electric current flow from Cathode to anode in external circuit whereas current flows from anode to cathode in internal circuit
- (ii) During cell reaction weight of anodic terminal decreases whereas weight of cathodic terminal increases
- (iii) During cell reaction concentration of ion in anodic solution increases whereas concentration of Cu^{2+} ion in cathodic solution decreases
- (iv) Cell reactions are spontaneous and exothermic therefore during cell reaction temperature of cell is increase.

SALT-BRIDGE

- (i) It is a glassed U-tube which is filled by inert compound like KCl , KNO_3 , NH_4NO_3 etc with gelating or agar-agar to convert into semi-solid form or past form by which rate of diffusion of ion decreases. Cotton plug is also plugged at the end of salt bridge
- (ii) It connect two different electrolytic solution and completes the cell circuit
- (iii) It maintain electrolytical neutrality of both electrolytic solution by supplying Cl^- in anodic and K^+ in cathodic
- (iv) It prevents liquid-liquid junction potential

asn. what is liquid-liquid junction potential

Ans when two different electrolytic solutions are placed in direct contact with each other then a junction of potential is developed which is called as liquid-liquid junction potential and its nature is always such that it opposes the e.m.f of cell

⇒ According to thermodynamics ΔG is equal to useful work by the system

$$\Delta G = -W_{\text{useful}} = -W_{\text{electrical}}$$

$$\Delta G = -Q dv = -n F \times E_{\text{cell}}$$

$$\boxed{\Delta G = -n F E_{\text{cell}}}$$

n = no of moles of transferred e^-

F = Faraday constant = 96500 C

$$\Delta G_{\text{cell}} = (\Delta G_{\text{oxidation}})_{\text{Anode}} + (\Delta G_{\text{reduction}})_{\text{cathode}}$$

$$-n F E_{\text{cell}} = (-n F E_{\text{oxi}})_{\text{Anode}} + (-n F E_{\text{red}})_{\text{c}}$$

$$-n F E_{\text{cell}} = -n F (E_{\text{ox}})_A + (-n F E_{\text{red}})_C$$

$$-n F E_{\text{cell}} = -n F (E_{\text{ox}})_A + (E_{\text{red}})_C$$

$$E_{\text{cell}} = (E_{\text{oxi}})_A - (E_{\text{red}})_C$$

$$E_{\text{cell}} = (E_{\text{red}})_C - (E_{\text{red}})_A$$

$$E_{\text{cell}} = (E_{\text{ox}})_A - (E_{\text{ox}})_C$$

°° SRP = -SOP

IMP ⇒ if conditions are standard

$$E^{\circ}_{\text{cell}} = (E^{\circ}_{\text{red}})_C - (E^{\circ}_{\text{oxi}})_A$$

$$\Delta G^{\circ} = -n F E_{\text{cell}}$$

⇒ if cell reaction is spontaneous = $\Delta G = -ve$ $E_{\text{cell}} = +ve$

⇒ if cell reaction is in equilibrium = $\Delta G = 0$ $E_{\text{cell}} = 0$

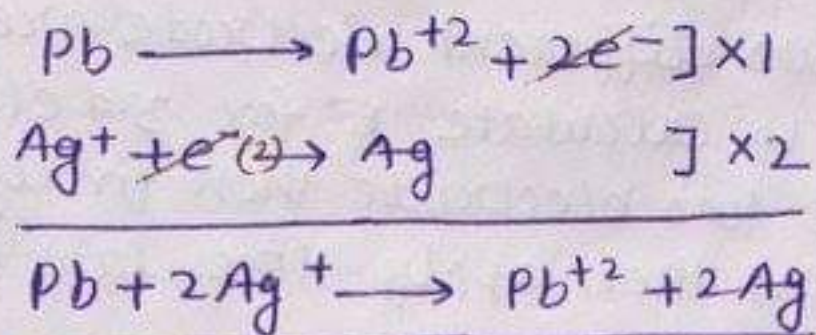
$\Delta G \neq 0$ $E_{\text{cell}} \neq 0$

KOTA CLASSNOTES

www.notesdrive.com

Qsn Pb rod was dipped in $Pb(NO_3)_2$ solution and Ag rod was dipped in $AgNO_3$ solution then write down cell reaction and cell representation

Ans $Pb \longrightarrow Pb(NO_3)_2 = \text{cathode}$
 $Ag \longrightarrow Ag(NO_3) = \text{anode}$



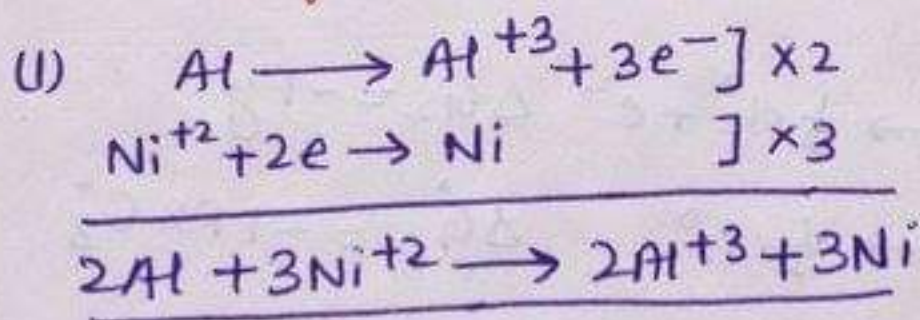
cell representation = $Pb/Pb^{+2} // Ag^+/Ag$

Qsn For the cell $Al/Al^{+3} // Ni^{+2}/Ni$

$$E^\circ_{Ni/Ni^{+2}} = +0.25 V$$

$$E^\circ_{Al^{+3}/Al} = -1.66 V$$

Then find out (i) cell reaction (ii) E°_{cell} (iii) ΔG°_{cell}



$$\begin{aligned} (ii) \quad E^\circ_{cell} &= (E^\circ_{red})_c - (E^\circ_{red})_A \\ &= -0.25 - (-1.66) \\ &= +1.41 \text{ Volt} \end{aligned}$$

$$\begin{aligned} (iii) \quad \Delta G^\circ &= -nF E^\circ_{cell} \\ \Delta G^\circ &= -6 \times 96500 \text{ Cb} \times 1.41 \text{ J} \\ \Delta G^\circ &= -816390 \text{ J} \end{aligned}$$

-ve sign of ΔG° shows that reaction is spontaneous

SPECIAL NOTES

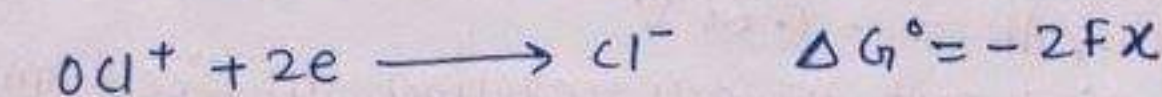
- * Value of electrode potential doesn't change when it is multiplied/divided by a coefficient because electrode potential is an intrinsic property
- * Value of electrode potential doesn't depend on size of metallic rod
- * If E° for two different electrodes are given and we want to calculate E° for 3rd electrode by using these two electrodes then firstly convert E° into ΔG° and then apply HEISS law of energetics

Qsn

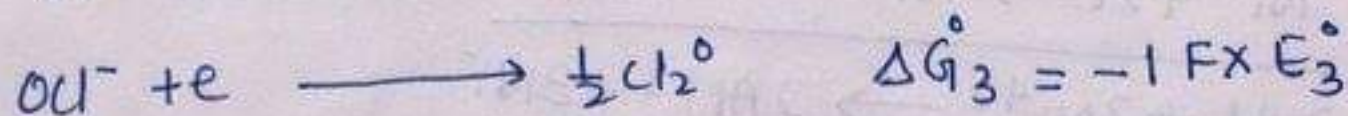
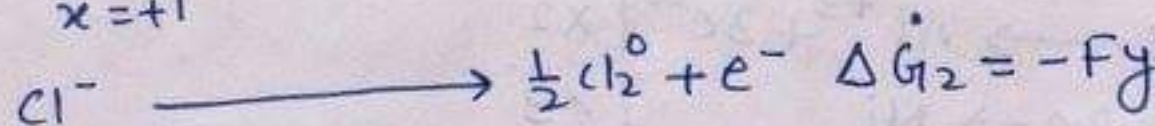
$$E^\circ \text{OCl}^- / \text{Cl}^- = x$$

$$E^\circ \text{Cl}^- / \frac{1}{2} \text{Cl}_2 = y$$

$$E^\circ \text{OCl}^- / \frac{1}{2} \text{Cl}_2 = ?$$

KOTA CLASSNOTESAns

$$\begin{aligned} -2 + x &= -1 \\ x &= +1 \end{aligned}$$



$$\text{Eqn (1)} + \text{Eqn (2)} = \text{Eqn (3)}$$

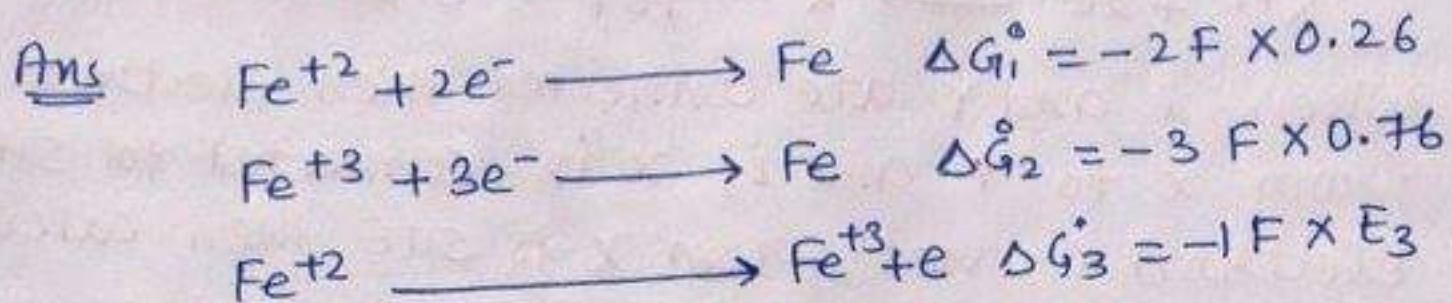
$$\Delta G^\circ + \Delta G^\circ_2 = \Delta G^\circ_3$$

$$-2Fx - Fy = -Fx E^\circ_3$$

$$(-2x - y) = -E^\circ_3$$

$$\underline{E^\circ = 2x + y}$$

Q.51 $E^\circ \text{ Fe}^{+2}/\text{Fe} = 0.26 \text{ V}$
 $E^\circ \text{ Fe}^{+3}/\text{Fe} = 0.76 \text{ V}$
 $E^\circ \text{ Fe}^{+2}/\text{Fe}^{+3} = ?$



$$+ \text{eqn (1)} - \text{eq (2)} = \text{eq (3)}$$

$$\Delta G_1^\circ - \Delta G_2^\circ = \Delta G_3^\circ$$

$$-0.52F + 2.28 = -F \times E_3$$

$$+1.76 = -E_3$$

$$\boxed{E_3 = -1.76}$$

HOME
WORK
VVI

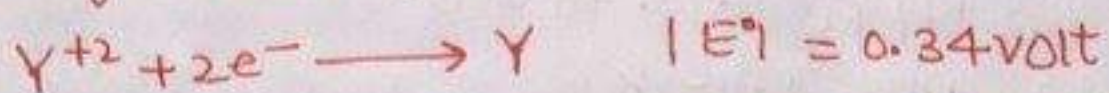
$$E^\circ \text{ Cu}^{+2}/\text{Cu}^+ = +0.15 \text{ V}$$

$$E^\circ \text{ Cu}^{+2}/\text{Cu} = +0.34 \text{ V}$$

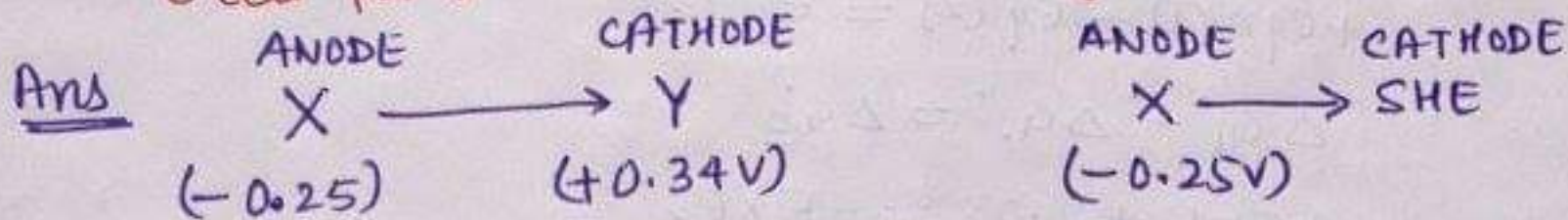
$$E^\circ \text{ for disproportionation of Cu}^+ = ?$$

KOTA CLASSNOTES

Qsn Magnitude of SRP of two metals x and y are



When x and y are connected then electron crossed from x to y and if x is connected to SHE then electron crossed from x to SHE then calculate E°_{cell} for the cell made up by x and y

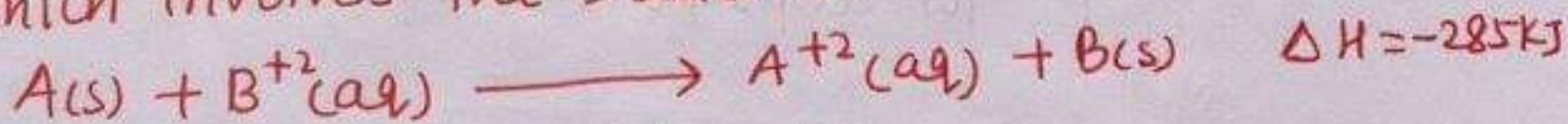


$$E^{\circ}_{\text{cell}} = (E^{\circ}_{\text{red}})_C - (E^{\circ}_{\text{red}})_A$$

$$E^{\circ}_{\text{cell}} = 0.34 + 0.25$$

$$\boxed{E^{\circ}_{\text{cell}} = 0.59}$$

Qsn The Efficiency of a hypothetical cell is 84%. which involves the reaction



Then calculate E° of cell

Ans

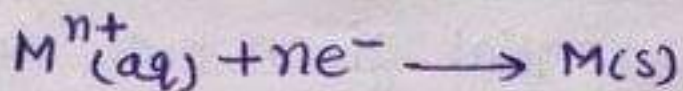
$$\text{Efficiency} = \frac{\Delta G^{\circ}}{\Delta H^{\circ}} = \frac{-nFE^{\circ}_{\text{cell}}}{\Delta H}$$

$$0.84 = \frac{-2 \times 96500 \times E^{\circ}_{\text{cell}}}{-285 \times 10^3}$$

$$E^{\circ}_{\text{cell}} = \frac{0.84 \times -285 \times 10^3}{-2 \times 96500}$$

NERST EQUATION

- * We have assumed in the previous section that the concentration of the species involved in the reaction is unity. This need not be always true.
- * Nerst showed that for the electrode reaction



- * The electrode potential at any concentration measured with respect SHE can be represented by

$$E = E^{\circ} - \frac{RT}{nF} \log_e \frac{[Product]}{[Reactant]}$$

E = Electrode potential (volt)

E° = Standard electrode potential

R = Gas constant ($8.314 \text{ J mol}^{-1} \text{ K}^{-1}$)

T = Temperature (kelvin)

n = number of moles of e^{-} transferred

F = Faraday ($96500 \text{ C/mol e}^{-}$)

$[]$ = Active mass or concentration

$$\frac{[P]}{[R]} = Q$$

Generally in question ($T = 298 \text{ K}$) given then:-

$$\frac{2.303 RT}{F} = \frac{2.303 \times 8.314 \text{ J mol}^{-1} \text{ K}^{-1} \times 298}{96500} = 0.591$$

$$E = E^{\circ} - \frac{0.591}{n} \log_{10} \frac{[P]}{[R]}$$

This formula valid only at 298 K

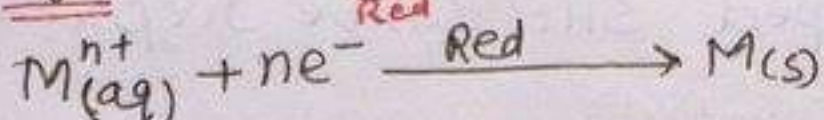
APPLICATION OF NERST EQUATION

- (i) To calculate electrode potential
- (ii) To calculate electrode potential of H_2 electrode
- (iii) To calculate e.m.f of cell

[I] TO CALCULATE ELECTRODE POTENTIAL

if only electrode potential given then take SRP

eg:- $E(M^{n+}/M) = ?$



$$E_{\text{red}} = E^{\circ}_{\text{red}} - \frac{0.591}{n} \log_{10} \frac{[M]}{[M^{n+}]}$$

1 Because solid

$$E_{\text{red}} = E^{\circ}_{\text{red}} + \frac{0.591}{n} \log [M^{n+}]$$

for Metal electrode

If $[M^{n+}] \uparrow \Rightarrow E_{\text{red}} \uparrow \Rightarrow E_{\text{oxi}} \downarrow$

If $[M^{n+}] \downarrow \Rightarrow E_{\text{red}} \downarrow \Rightarrow E_{\text{oxi}} \uparrow$

Q.51 $E(M/M^{n+}) = ?$

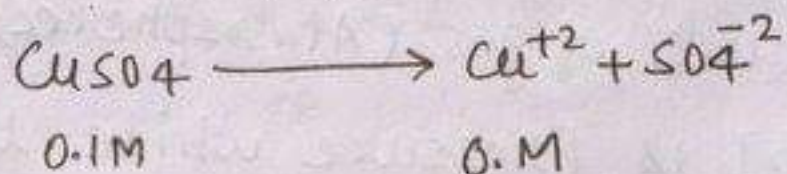


Here oxidation so take formula of oxidation

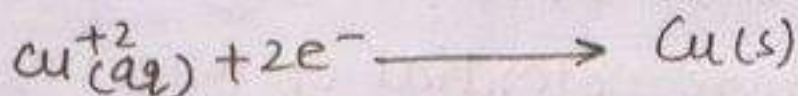
$$E_{\text{oxi}} = E^{\circ}_{\text{oxi}} - \frac{0.0591}{n} \log_{10} \frac{[M^{n+}]}{[M]} \rightarrow 1 \because \text{for solid active mass} = 1$$

Qsn Calculate electrode potential of a copper electrode dipped in a 0.1M solution of CuSO_4 at 25°C . The standard electrode potential Cu^{2+}/Cu is 0.34 volt at 298 K

Ans CuSO_4 is a salt and α is not given means 100% ionisation occurs



$$[\text{Cu}^{2+}] = C\alpha = 0.1 \times \frac{100}{100} = 0.1$$



$$E_{\text{red}} = E^\circ_{\text{red}} - \frac{0.0591}{2} \log_{10} \frac{[\text{Cu}]}{[\text{Cu}^{2+}]}$$

1 (∵ for solid active mass is 1)

$$= 0.34 - \frac{0.0591}{2} \log_{10} \frac{1}{0.1} = 0.34 - 0.03 = 0.31 \text{ volt}$$

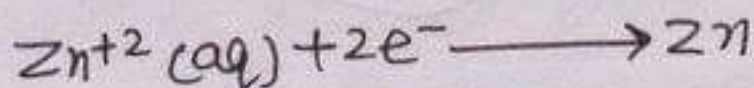
Qsn A piece of Zn metal is dipped in a 0.1M solution of Zn salt. The salt is dissociated to the extent of 20%. Calculate the electrode potential of Zn^{2+}/Zn .

(Given $E^\circ(\text{Zn}^{2+}/\text{Zn}) = -0.76 \text{ volt}$)

Here α is given so salt will not dissociate 100%. It will dissociate only up to 20%.

$$C = 0.1\text{M} \quad [\text{Zn}^{2+}] = C\alpha$$

$$[\text{Zn}^{2+}] = 0.1 \times 20\% = 0.1 \times \frac{20}{100} = 0.02\text{M}$$



$$E_{\text{red}} = E^\circ_{\text{red}} - \frac{0.0591}{n} \log_{10} \frac{[\text{Product}]}{[\text{Reactant}]}$$

$$= -0.76 - \frac{0.0591}{2} \log_{10} \frac{[\text{Zn}]}{[\text{Zn}^{2+}]}$$

1 (∵ for solid active mass = 1)

$$= -0.76 - 0.03 \log_{10} \frac{1}{0.02}$$

$$= -0.76 - 0.03 \log_{10} \left(\frac{100}{2} \right) \Rightarrow -0.811 \text{ volt}$$

(III) TO CALCULATE ELECT. POTENTIAL OF H₂ ELECTRODE.

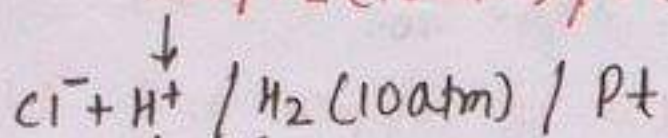
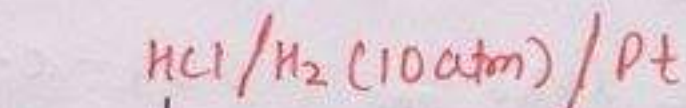


$$E_{red} = E^{\circ}_{red} - \frac{0.0591}{2} \log \frac{[PH_2]}{[H^+]^2}$$

* In Nernst equation concentration term is replaced by partial pressure of gas (Atmospheric)

* Here $\log [PH_2]$ is pressure while $\log \frac{1}{[H^+]^2}$ is concentration मगर ये गलत नहीं है। ये फिर भी सही है क्योंकि $\log$ एक unitless है। इसे replace किया जा सकता है।

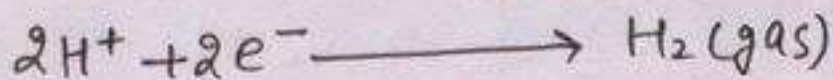
Qsn Calculate electrode potential of following electrode when pH of acidic solution is 2



Reduction

$$[H^+] = 10^{-pH} = [10^{-2}]$$

KOTA CLASSNOTES



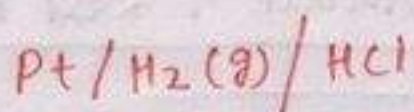
$$E_{red} = E^{\circ}_{red} - \frac{0.0591}{2} \log \frac{[PH_2]}{[H^+]^2}$$

$$E_{red} = 0 - \frac{0.06}{2} \log \left(\frac{[10]}{[10^{-2}]^2} \right) 10^5$$

$$= -0.03 \times 5$$

$$= \underline{\underline{-0.15 \text{ volt}}}$$

Qsn calculate pH of following half cell reaction
Given ($E_{oxi} = 0.25V$)



Ans



$$E_{oxi} = E_{oxi} - \frac{0.0591}{2} \log \frac{[H^+]^2}{[P_{H_2}]}$$

$$E_{oxi} = - \frac{0.0591}{2} \times 2 \log [H^+]$$

$$\frac{E_{oxi}}{0.0591} = - \log [H^+]$$

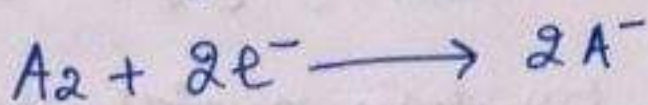
$$pH = \frac{E_{oxi}}{0.0591} = \frac{-E_{red}}{0.0591}$$

$$pH = \frac{0.25}{0.0591} = 4.16$$

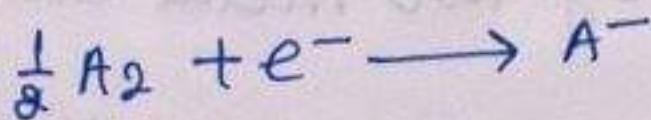
KOTA CLASSNOTES

NOTE :

- (i) Value of electrode potential for a given electrode doesnot change when reaction is multiplied or divided by coefficient



$$E = x \text{ volt}$$



$$E = ?$$

$$\text{Ans} = x \text{ volt } (*: \text{No change})$$

- (ii) Value of electrode potential doesnot depends on size of electrode and amount of solution but depends on concentration of solution. It is an intensive property

III] TO CALCULATE E.M.F OF THE CELL

$$E_{\text{cell}} = \text{Oxi potential of Anode} + \text{Red pote of Cathode}$$

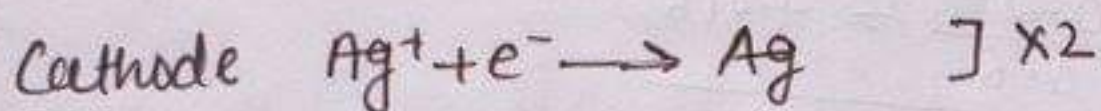
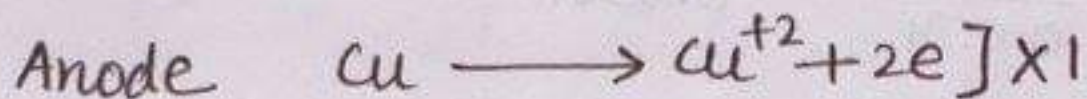
$$E_{\text{cell}} = E_{\text{cathode}} - E_{\text{anode}}$$

At Standard condition

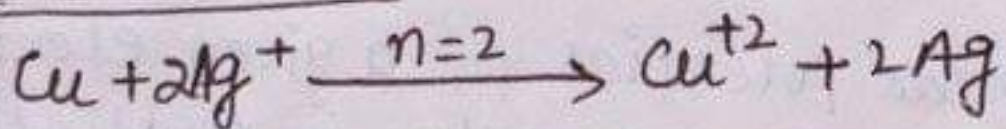
$$E^{\circ}_{\text{cell}} = E^{\circ}_{\text{C}} - E^{\circ}_{\text{A}}$$

Stand. Reduction potential

eg $1 \times (\text{Cu} / \text{Cu}^{+2}(\text{aq})) // (\text{Ag}^{+}(\text{aq}) / \text{Ag}) \times 2 \quad E_{\text{cell}} = ?$



Cell reaction



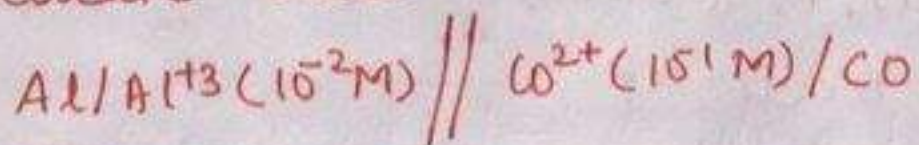
$$E_{\text{cell}} = E^{\circ}_{\text{cell}} - \frac{0.0591}{n} \log \frac{[\text{Cu}^{+2}] [\text{Ag}]^2}{[\text{Ag}^{+}]^2 [\text{Cu}]}$$

KOTA CLASSNOTES

When cell is formed by two metal electrodes

$$E_{\text{cell}} = E^{\circ}_{\text{cell}} - \frac{0.0591}{n} \log \frac{[\text{Anode}]^x}{[\text{Cathode}]^y}$$

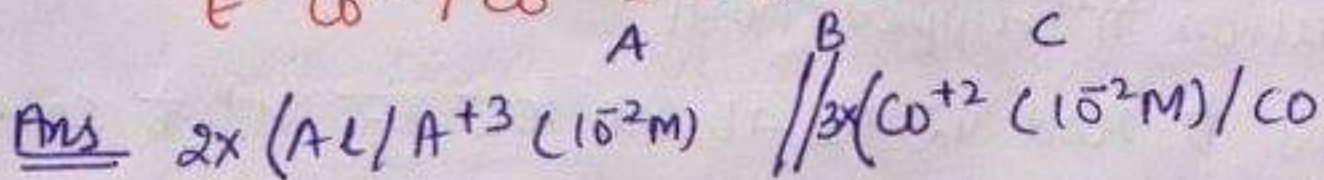
Qsn Calculate $E_{\text{cell}} = ?$



Given

$$E^\circ \text{Al}^{+3} / \text{Al} = -1.66 \text{V}$$

$$E^\circ \text{Co}^{2+} / \text{Co} = -0.28 \text{V}$$



$$2 \times 3e^- = 6e^-$$

$$3 \times 2e^- = 6e^- \quad n = 6e^-$$

$$E^\circ \text{Al}^{+3} / \text{Al} = -1.66 \text{V},$$

Red

$$E^\circ \text{Co}^{2+} / \text{Co} = -0.28 \text{V}$$

Red

$$E^\circ_{\text{cell}} = E^\circ_{\text{C}} - E^\circ_{\text{A}}$$

Anode हो या cathode हो दोनों की SRP put करनी है।

$$E^\circ_{\text{cell}} = (\text{Co}) - (\text{Al})$$

$$E^\circ_{\text{cell}} = -0.28 - (-1.66)$$

$$E^\circ_{\text{cell}} = 1.38 \text{V}$$

KOTA CLASSNOTES

$$E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{0.0591}{6} \log \frac{[\text{Al}^{+3}]^2}{[\text{Co}^{2+}]^3}$$

$$= 1.38 - 0.01 \log \frac{[10^{-2}]^2}{[10^{-1}]^3}$$

$$= 1.38 - 0.01 \log 10^{-1}$$

$$= 1.38 + 0.01$$

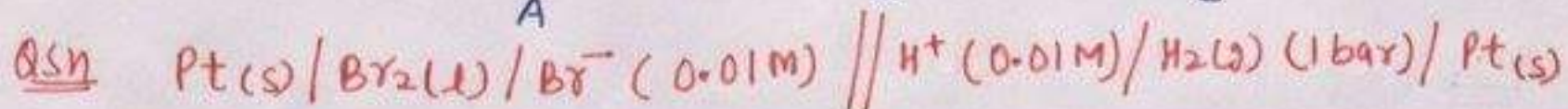
$$= 1.39 \text{ volt}$$

(Anode)

(Bridge)

(Cathode)

44



What is $E_{\text{cell}} = ?$

Given - ($E^\circ_{\text{Pt}/\text{Br}_2/\text{Br}^-} = 1.09\text{V}$)

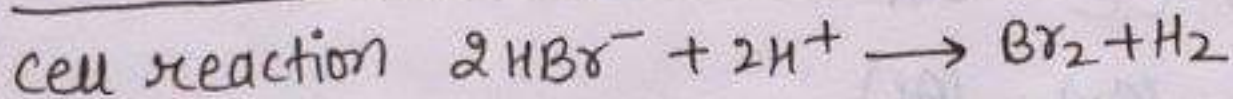
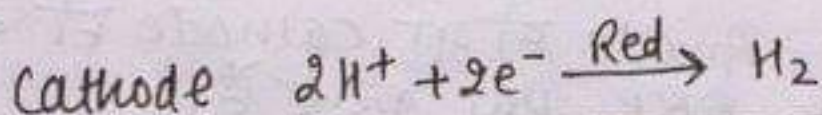
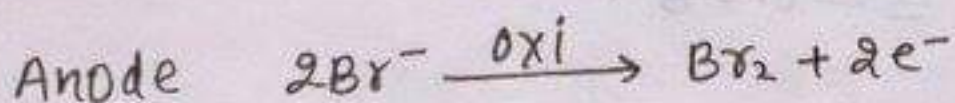
Ans * Pt(s) के mod पर Br^- को absorb करा कर Br^- के solution में dipped किया

* Pt(s) के mod पर H_2 absorb करा कर H^+ के solution में dipped किया

$$E^\circ_{\text{cell}} = E^\circ_{\text{C}} - E^\circ_{\text{A}}$$

$$= 0 - 1.09$$

$$= -1.09\text{V}$$



$$E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{0.0591}{2} \log \frac{[\text{Br}_2] [\text{PH}_2]}{[\text{Br}^-]^2 [\text{H}^+]^2}$$

$$= -1.09 - 0.03 \log \left(\frac{1}{[10^{-2}]^2 [10^{-2}]^2} \right) - 10^{-8}$$

$$= -1.09 - 0.03 \times 8$$

$$= -1.33\text{V}$$

KOTA CLASSNOTES

www.notesdrive.com

$\Rightarrow$ Work done by a system = $\downarrow$ in Free energy of system ($-\Delta G$)

$\Rightarrow$ Useful work obtained by cell = in the form of electrical energy

$$W = Q \times V$$

$$\boxed{W = nF \times E_{\text{cell}}}$$

$$\Rightarrow W = nF E_{\text{cell}} = -\Delta G$$

$$\Rightarrow \boxed{\Delta G = -nFE} \quad n = \text{no of moles of transferred electrons}$$

* At Standard Condition

$$\boxed{\Delta G^\circ = -nFE^\circ_{\text{cell}}}$$

CASE - I

If $E_{\text{cell}} = +ve$ OR $\Delta G = -ve$

then given cell reaction is spontaneous

CASE - II

If $E_{\text{cell}} = -ve$ OR $\Delta G = +ve$

Then given cell reaction is non-spontaneous But
It will be spontaneous in reverse condition

CASE - III

If $E_{\text{cell}} = 0$ OR $\Delta G = 0$

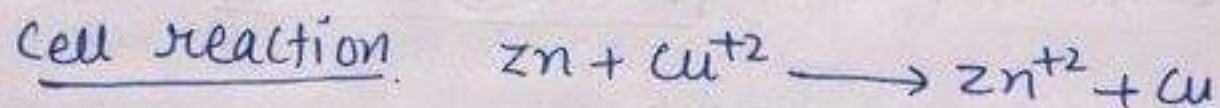
Then given cell reaction will be in equilibrium

Qsn find spontaneity of reaction in Daniel cell reaction
 $\text{Zn} / \text{Zn}^{2+} (1\text{M}) \parallel \text{Cu}^{2+} (1\text{M}) / \text{Cu}$

Given $E^\circ \text{Zn}^{2+} / \text{Zn} = -0.76 \text{ volt}$

$E^\circ \text{Cu}^{2+} / \text{Cu} = 0.34 \text{ volt}$

Ans



$$E^\circ_{\text{cell}} = E^\circ_{\text{c}} - E^\circ_{\text{a}}$$

$$= (\text{Cu}) - (\text{Zn})$$

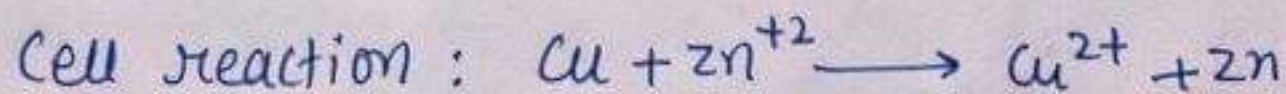
$$= 0.34 - (-0.76\text{v})$$

$E^\circ_{\text{cell}} = +1.1 \text{ volt}$ OR $\Delta G = -ve$

$E^\circ_{\text{cell}} = +ve$ means spontaneous

Qsn $\text{Cu} / \text{Cu}^{2+} (1\text{M}) \parallel \text{Zn}^{2+} (1\text{M}) / \text{Zn}$

Ans



$$E^\circ_{\text{cell}} = E^\circ_{\text{c}} - E^\circ_{\text{a}}$$

$$(\text{Zn}) \quad (\text{Cu})$$

$$= -0.76 - (+0.34)$$

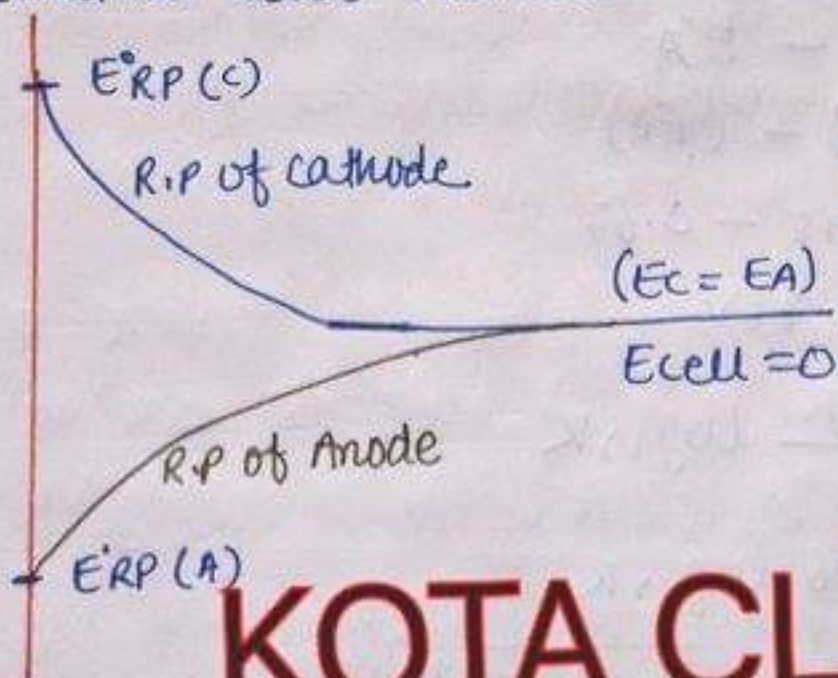
$E^\circ_{\text{cell}} = -1.1 \text{ volt}$ OR $\Delta G = +ve$

$E^\circ_{\text{cell}} = -ve$ means non-spontaneous.

KOTA CLASSNOTES

CONCEPT OF EQUILLIBRIUM

When cell reaction occurs concentration of ions in Cathode solution decreases therefore its reduction Potential also decreases but concentration of ions in Anode solution increases therefore its reduction Potential also increases.



$$* E^{\circ}_{cell} = E_{cath} - E_{anode}$$

$$E^{\circ}_{cell} = 0$$

$$\Delta G = 0$$

* $\Delta G = 0$ cell rxn is equilib

KOTA CLASSNOTES

According to Nernst equation

$$E_{cell} = E^{\circ}_{cell} - \frac{RT}{nF} \ln \frac{[P]}{[R]} = \frac{[Anode] \uparrow}{[Cathode] \downarrow} \Rightarrow Q$$

At equilibrium $E_{cell} = 0$, $Q = K$ (eqm constt for cell rxn)

$$E_{cell} = E^{\circ}_{cell} - \frac{RT}{nF} \ln K$$

$$0 = E^{\circ}_{cell} - \frac{RT}{nF} \ln K$$

$$E^{\circ}_{cell} = \frac{RT}{nF} \log_{10} K$$

$$E^{\circ}_{cell} = \frac{0.0591}{n} \log_{10} K$$

$$\Delta G^{\circ} = -nFE^{\circ}$$

$$\Delta G^{\circ} = -nF \left\{ \frac{RT}{nF} \log_{10} K \right\}$$

$$\Delta G^{\circ} = -2.303 RT \log_{10} K$$

Qsn Calculate equilibrium constant for the reaction



Given $\Rightarrow E^\circ_{\text{Ce}^{4+}/\text{Ce}^{3+}} = 1.45\text{V}$

$E^\circ_{\text{Fe}^{3+}/\text{Fe}^{2+}} = 0.67\text{V}$

Ans

$$\begin{aligned} E^\circ_{\text{cell}} &= E^\circ_{\text{c}} - E^\circ_{\text{a}} \\ &= (\text{Ce}) - (\text{Fe}) \\ &= 1.45 - 0.67 \\ &= 0.78\text{V} \end{aligned}$$

$$E^\circ_{\text{cell}} = \frac{0.0591}{1} \log_{10} K$$

$$0.78 = 0.06 \log_{10} K$$

$$\log_{10} K = 13$$

$$K = 10^{13}$$

Qsn In which direction the reaction is proceed spontaneous at following concentration of ion participating in rxn



(i) $[\text{Ag}^+] = 10^{-4}\text{M}$; $[\text{Hg}_2^{+2}] = 10^{-1}\text{M}$

(ii) $[\text{Ag}^+] = 10^{-1}\text{M}$; $[\text{Hg}_2^{+2}] = 10^{-4}\text{M}$

Given

$$E^\circ_{\text{Hg}_2^{+2}/\text{Hg}} = 0.79\text{V}$$

$$E^\circ_{\text{Ag}^+/\text{Ag}} = 0.80\text{V}$$

Ans

$$E^\circ_{\text{cell}} = E^\circ_{\text{c}} - E^\circ_{\text{a}}$$

$$E^\circ_{\text{cell}} = 0.80 - 0.79$$

$$E^\circ_{\text{cell}} = 0.1\text{V}$$

$$\begin{aligned} E_{\text{cell}} &= E^\circ_{\text{cell}} - \frac{0.0591}{2} \log \frac{[\text{Hg}_2^{+2}]}{[\text{Ag}^+]^2} \\ &= 0.01 - 0.03 \log \left(\frac{10^{-1}}{(10^{-4})^2} \right) \Rightarrow 10^{-7} \\ &= 0.01 - 0.03 \times 7 = -0.20\text{V} \end{aligned}$$

(i) $E_{\text{cell}} = -0.20\text{V}$; $\Delta G = +ve$ (Right to Left)

(ii) $E_{\text{cell}} = 0.07\text{V}$; $\Delta G = -ve$ (Left to Right)

Q.57 Given standard electrode potential in 1M acidic solⁿ

$$E^\circ (\text{Fe}^{2+}/\text{Fe}) = -0.4\text{V}$$

$$E^\circ (\text{Fe}^{3+}/\text{Fe}^{2+}) = 0.8\text{V}$$

$$E^\circ (\text{Mn}^{2+}/\text{Mn}) = -1.2\text{V}$$

$$E^\circ (\text{Mn}^{3+}/\text{Mn}^{2+}) = 1.5\text{V}$$

(i) Comment on relative stability of +2 and +3 oxidation state of Iron and manganese

(ii) which of the two metal is more easily to oxidise (+2) state

Ans * $E^\circ (\text{Fe}^{2+}/\text{Fe}) = -0.4\text{V}$

KOTA CLASSNOTES

$E^\circ = -ve$ and $\Delta G = +ve$ that's why

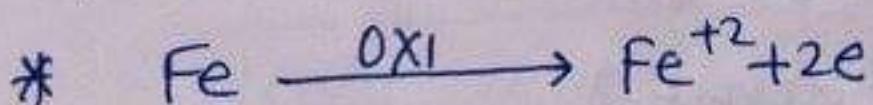
Fe^{+2} को Fe में नहीं जाना चाहेगी बल्कि Fe को Fe^{2+} में जाना चाहेगी

* $E^\circ (\text{Mn}^{2+}/\text{Mn}) = -1.2\text{V}$

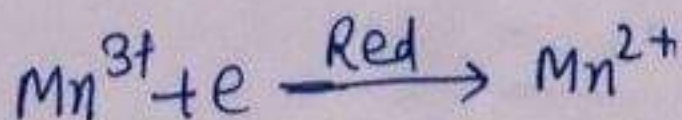
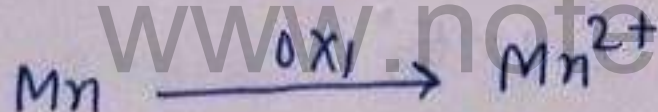
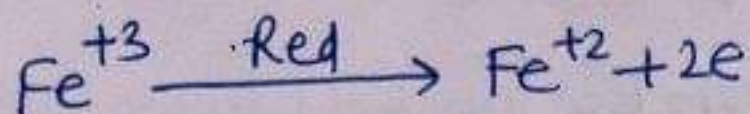
$E^\circ = -ve$ and $\Delta G = +ve$ that's why

Mn^{2+} को Mn में नहीं जाना चाहेगी But Mn तो Mn^{2+} में जाना चाहेगी

$\Rightarrow \text{Fe(II)} > \text{Fe(III)}$
 $\text{Mn(II)} > \text{Mn(III)}$



$E^\circ_{\text{ox}} = +0.4$
 $\Delta G^\circ_{\text{ox}} = -ve$ } spontaneous



Qsn $E^\circ (\text{Fe}^{3+}/\text{Fe}^{2+}) = ?$

Given $E^\circ (\text{Fe}^{3+}/\text{Fe}) = -0.036\text{V}$

$E^\circ (\text{Fe}^{2+}/\text{Fe}) = -0.440\text{V}$

Ans $\text{Fe}^{3+} + 3e^- \longrightarrow \text{Fe} \quad E_1^\circ = -0.036\text{V}$

$\text{Fe}^{2+} + 2e^- \longrightarrow \text{Fe} \quad E_2^\circ = -0.440\text{V}$

$\text{Fe}^{3+} + e^- \longrightarrow \text{Fe}^{2+}; E_3 = ?$

Funda:

KOTA CLASSNOTES

Since E° is an intrinsic property इसलिए पहले हम E° को ΔG° में convert करेंगे क्योंकि ΔG° is an extensive property means ΔG° is additive. Now solve the equation

$\text{Fe}^{3+} + 3e^- \longrightarrow \text{Fe} \quad E_1^\circ = -0.036\text{V}$

$\text{Fe}^{2+} + 2e^- \longrightarrow \text{Fe} \quad E_2^\circ = -0.440\text{V}$

$\text{Fe}^{3+} + e^- \longrightarrow \text{Fe}^{2+} \quad E_3 = ?$

$$\Delta G_1^\circ = -n_1 F E_1^\circ$$

$$\Delta G_2^\circ = -n_2 F E_2^\circ$$

$$\Delta G_3^\circ = -n_3 F E_3^\circ$$

Equation ① - equation ② = equation ③

$E_1^\circ - E_2^\circ = E_3^\circ \Rightarrow \times$ Because E° is an intensive (non-additive)

$$\Delta G_1^\circ - \Delta G_2^\circ = \Delta G_3$$

$$-n_1 F E_1^\circ - (-n_2 F E_2^\circ) = -n_3 F E_3^\circ$$

$$- \cancel{F} (n_1 E_1^\circ - n_2 E_2^\circ) = - \cancel{F} (n_3 E_3^\circ)$$

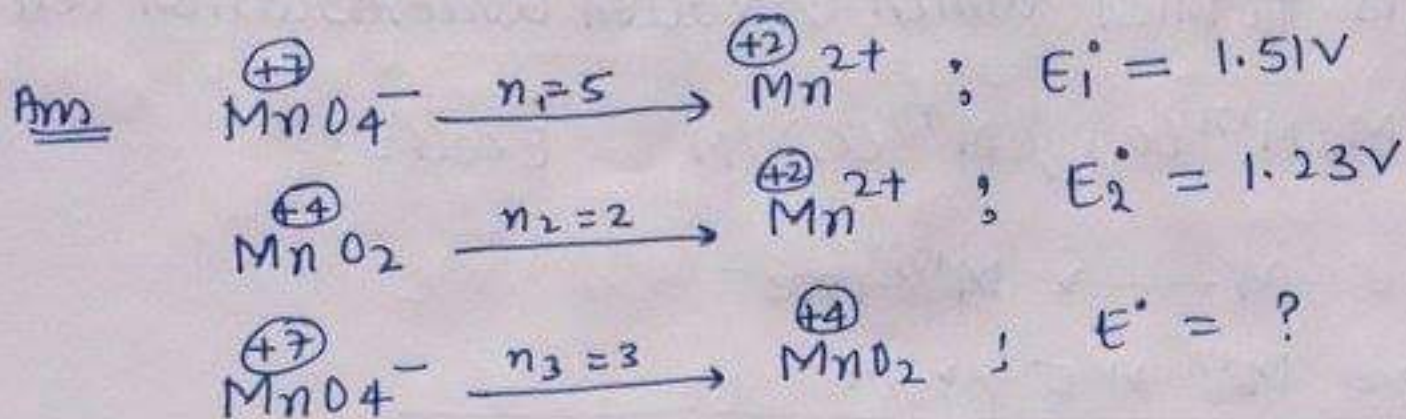
$$n_1 E_1^\circ - n_2 E_2^\circ = n_3 E_3^\circ$$

$$3 \times 0.036 - 2(-0.440) = 1 \times E_3^\circ$$

$$\underline{E_3^\circ = 0.772 \text{ volt}}$$

Q51 $E^\circ (\text{MnO}_4^- / \text{MnO}_2) = ?$

Given $E^\circ (\text{MnO}_4^- / \text{Mn}^{2+}) = 1.51\text{V}$
 $E^\circ (\text{MnO}_2 / \text{Mn}^{2+}) = 1.23\text{V}$



$$\text{Eqn (1)} - \text{Eqn (2)} = \text{Eqn (3)}$$

$$n_1 E_1^\circ - n_2 E_2^\circ = n_3 E_3^\circ$$

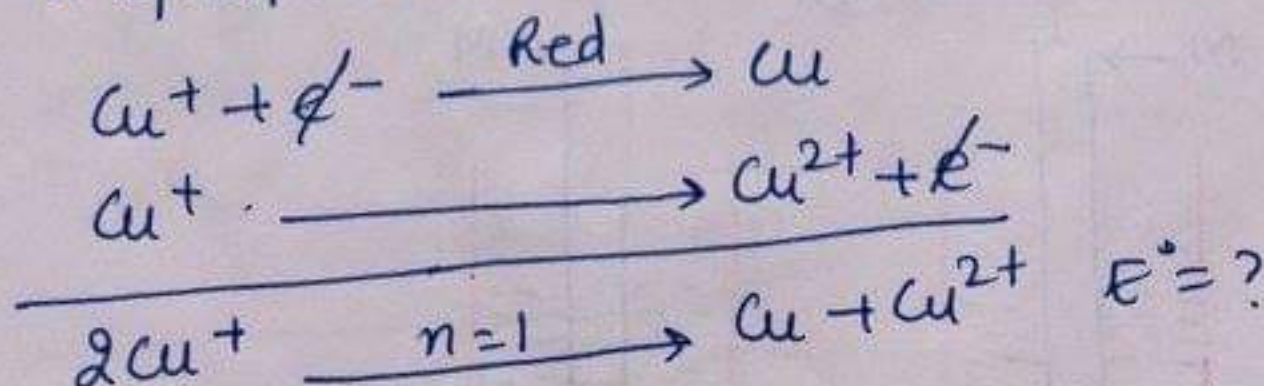
$$E_3^\circ = 1.69 \text{ volt}$$

KOTA CLASSNOTES

Q52 Cu^+ ion not stable in aqueous solution because of disproportionation reaction. Calculate E° for dispropo reaction.

Given $E^\circ (\text{Cu}^{2+} / \text{Cu}^+) = 0.15\text{V}$
 $E^\circ (\text{Cu}^{2+} / \text{Cu}) = 0.34\text{V}$

Ans Disproportionation of Cu



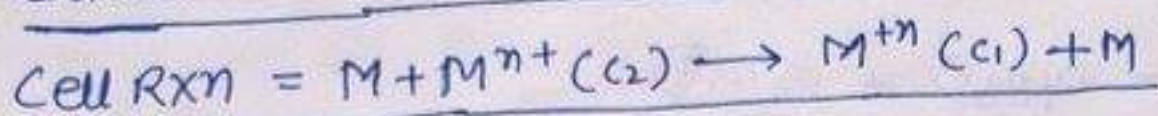
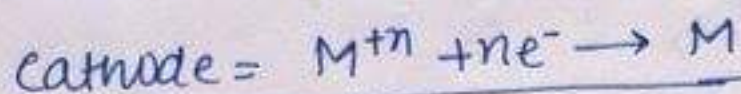
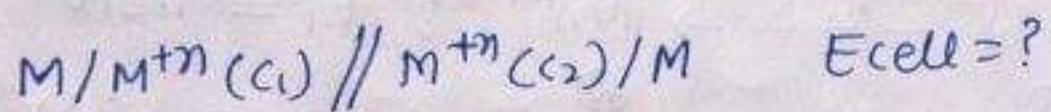
$$- 2 \times \text{eqn (1)} + \text{eqn (2)} = \text{eqn (3)}$$

$$- 2 \times n_1 E_1^\circ + n_2 E_2^\circ = n_3 E_3^\circ$$

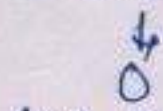
$$E_3^\circ = 0.38\text{V}$$

CONCENTRATION CELL

When two similar electrode having difference in concentration of ion in solution are combined then a cell is formed which is called concentration cell.



$$E_{\text{cell}} = E^{\circ}_{\text{cell}} - \frac{0.0591}{n} \log_{10} \left(\frac{c_1}{c_2} \right)$$



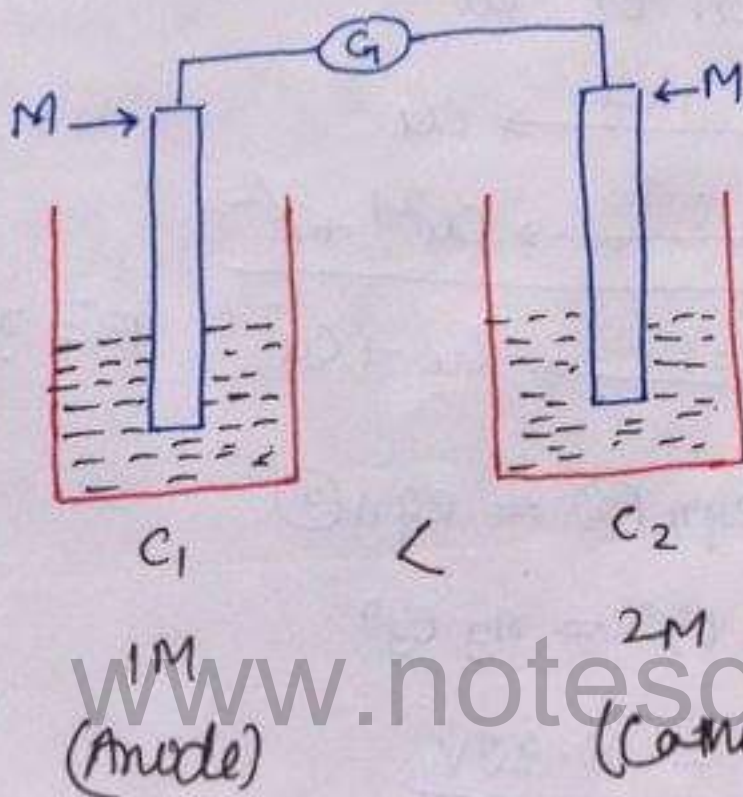
$$(E^{\circ}_{\text{cell}} = E_c - E_a)$$

KOTA CLASSNOTES

E°_{cell} is always zero (0) for concentration cell

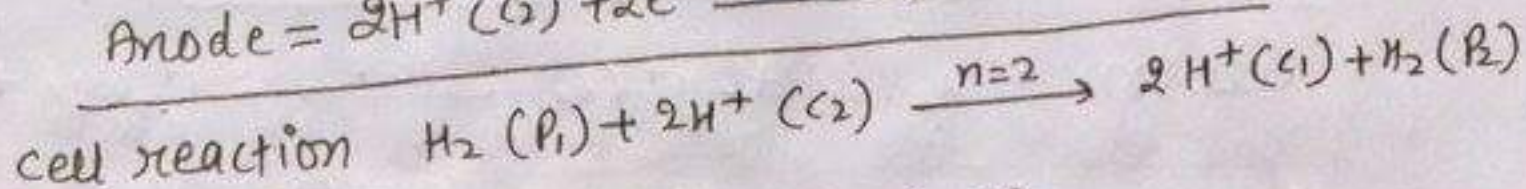
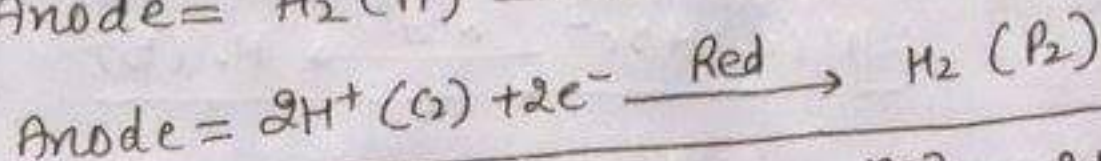
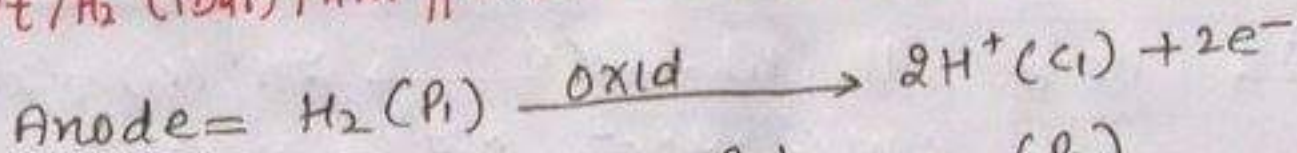
$$E_{\text{cell}} = \frac{+0.0591}{n} \log_{10} \left(\frac{c_2}{c_1} \right)$$

If $c_2 > c_1$ then $E_{\text{cell}} = +ve$ } spontaneous reaction
 $\Delta G = -ve$



QSN $\text{Pt} / \text{H}_2 (1 \text{ bar}) / \text{HA}_1 // \text{HA}_2 / \text{H}_2 (1 \text{ bar}) / \text{Pt} ; E_{\text{cell}} = ?$

Ans



$$E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{0.0591}{2} \log_{10} \frac{(\text{C}_1)^2}{(\text{C}_2)^2}$$

E_{cell}° for concentration cell = 0

$$E_{\text{cell}} = -\frac{0.0591}{2} \log \left(\frac{\text{C}_1}{\text{C}_2} \right)$$

$$E_{\text{cell}} = -0.0591 \log \left(\frac{(10^{-5})}{(10^{-3})} \right) \Rightarrow 10^{-2}$$

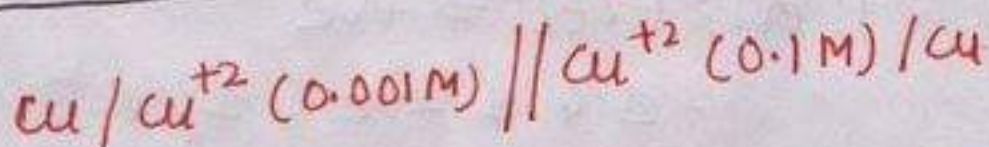
$$E_{\text{cell}} = +2 \times 0.0591$$

$$E_{\text{cell}} = 0.1182 \text{ V}$$

KOTA CLASSNOTES

AIIMS QSN
H.W

Hints



Concentration cell so $E^{\circ}_{\text{cell}} = 0$

$$E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{0.0591}{n} \log \left(\frac{\text{C}_1}{\text{C}_2} \right)$$

$$E_{\text{cell}} = -\frac{RT}{2F} \ln \left(\frac{10^{-3}}{10^{-1}} \right) \Rightarrow 10^{-2}$$

$$E_{\text{cell}} = -\frac{RT}{F} \ln 0.1$$

Q.57 $\text{Pt} / \text{H}_2(\text{g}) / \text{H}^+(\text{C}_1) // \text{H}^+(\text{C}_2) / \text{H}_2(\text{g}) / \text{Pt} ; E_{\text{cell}} = ?$

Ans Anode $\text{H}_2(\text{P}_1) \xrightarrow{\text{Oxid}} 2\text{H}^+(\text{C}_1) + 2\text{e}^-$
 $2\text{H}^+(\text{C}_2) + 2\text{e}^- \xrightarrow{\text{Red}} \text{H}_2(\text{P}_2)$

Cell rxn $= \text{H}_2(\text{P}_1) + 2\text{H}^+(\text{C}_2) \xrightarrow{n=2} 2\text{H}^+(\text{C}_1) + \text{H}_2(\text{P}_2)$

$$E_{\text{cell}} = E^{\circ}_{\text{cell}} - \frac{0.0591}{2} \log \frac{(\text{C}_1)^2 (\text{P}_2)}{(\text{C}_2)^2 (\text{P}_1)} \quad \text{--- (1)}$$

CASE-I $\text{P}_1 = \text{P}_2 = 1 \text{ bar}$ but $\text{C}_1 \neq \text{C}_2$

$$E_{\text{cell}} = 0 - \frac{0.0591}{2} \log \frac{\text{C}_1^2}{\text{C}_2^2}$$

CASE-II if $\text{C}_1 = \text{C}_2 = 1 \text{ M}$ but $\text{P}_1 \neq \text{P}_2$

$$E_{\text{cell}} = 0 - \frac{0.0591}{2} \log \left(\frac{\text{P}_1}{\text{P}_2} \right)$$

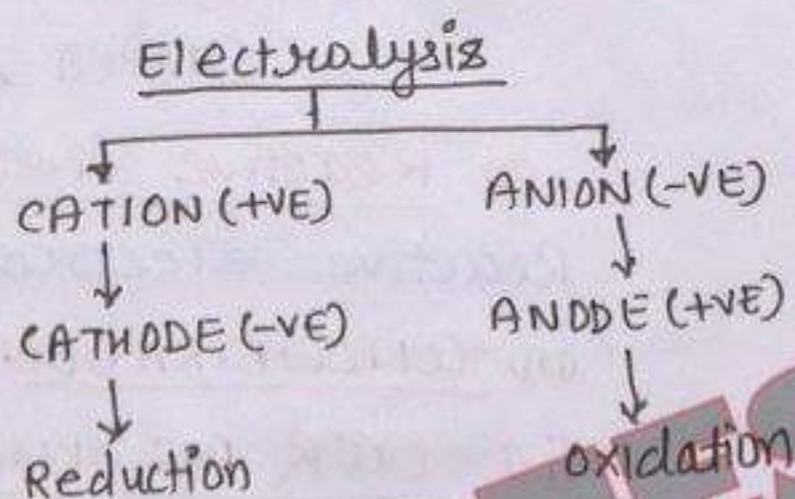
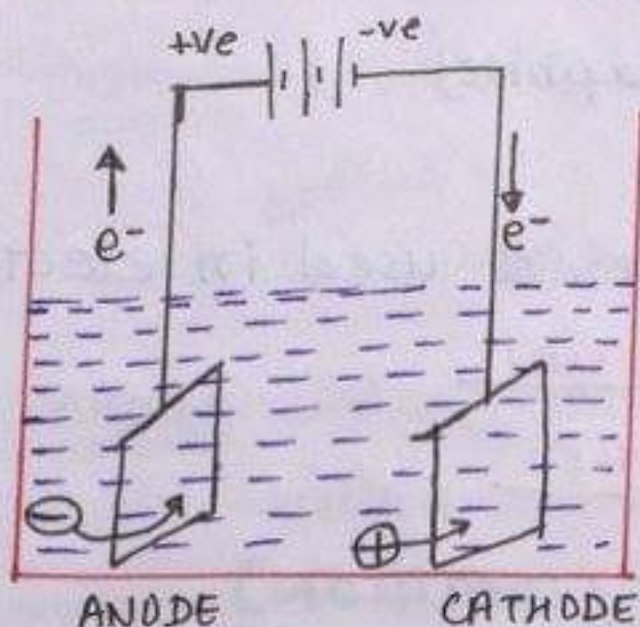
if $\text{P}_1 > \text{P}_2$ then $E_{\text{cell}} = +\text{ve}$
 $\Delta G = -\text{ve}$ spontaneous

PART TWO COMPLETES HERE

KOTA CLASSNOTES

PART- THREE - ELECTROLYSIS

- * The process of decomposition of an electrolysis when electricity passed through its aqueous solution or in molten state called electrolysis
- * The device in which electrolysis is carried out is called electrolytic cell



TRICK An OX (एक बैल)
Red Cat (लाल बिल्ली)

- * Sign of Anode and Cathode are opposite to Galvanic cell

* For Galvanic cell

$$\Delta G = -ve \text{ (spontaneous)}$$

Chem. energy $\rightarrow$ Elect. energy

For electrolytic cell

$$\Delta G = +ve \text{ (Non-spontaneous)}$$

Elect. energy $\rightarrow$ chemical energy

- * For electrolysis D.C is used क्योंकि जब A.C Pass करेंगे तो Electrode की Polarity change होता रहेगा. और Ion Confuse हो जाएगा कि Cathode पर जाए कि ANODE पर जाए.

- * To measure resistance of electrochemical cell we will use A.C क्योंकि D.C जब Pass करेंगे तो electrolysis होगा और अगर NaCl electrolysis करेगा तो NaOH बन जाएगा

- * A.C Pass करने पर Ion confusion में रहेगा और Cathode and anode के बीच combine हो जाएगा और हम easily electrolytic cell की resistance measure कर लेंगे.

Product of electrolysis mainly depends upon

(i) NATURE OF ELECTROLYTE (STRONG OR WEAK)

(ii) NATURE OF ELECTRODE

(A) inert (non-attacked) electrode

Inert electrode does not react with solution of electrolytes or with product of electrolysis. Such electrodes are used for electrolysis.

eg CHAPPA, C (graphite)

(B) Reactive electrodes

Reactive electrodes are not used in electrolysis

(iii) CONCENTRATION OF SOLUTION

(iv) OVER POTENTIAL OF GASES

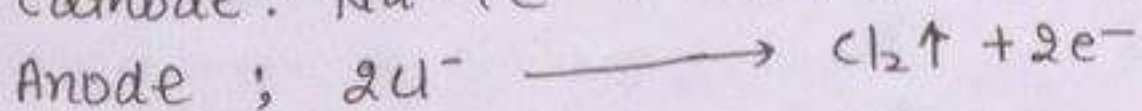
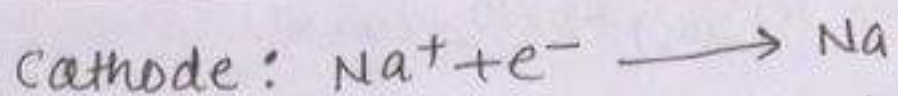
Eg OF ELECTROLYSIS (COMMON)

1. Electrolysis of fused (molten) NaCl using Pt electrode
2. Electrolysis of Aqueous solution of NaCl using Pt electrode
3. Electrolysis of Aqueous solution of CuSO_4 using Pt electrode
4. Electrolysis of Aqueous solution of AgNO_3 using Pt electrode
5. Electrolysis of Aqueous solution of Na_2SO_4 using Pt electrode
6. Electrolysis of Aqueous solution of NiSO_4 using Pt electrode

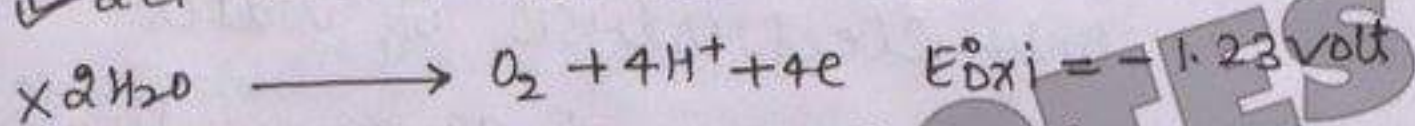
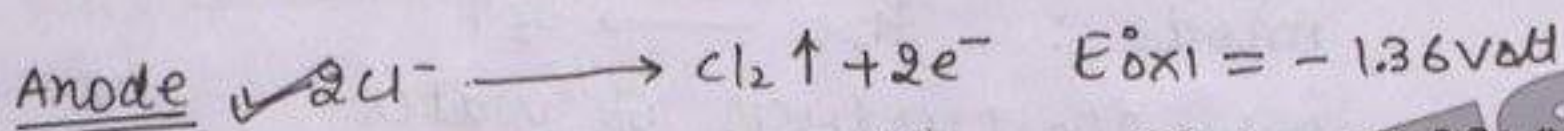
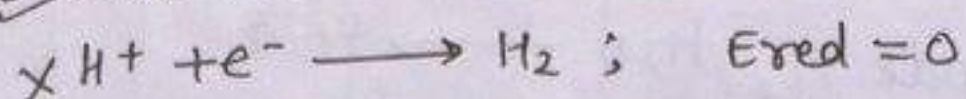
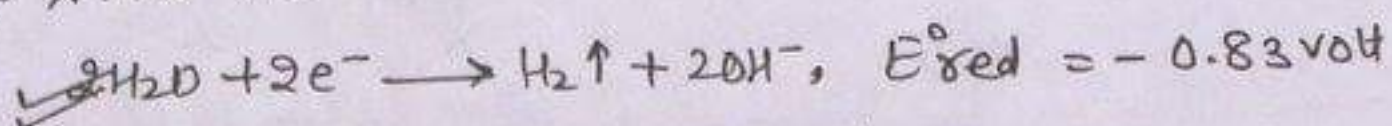
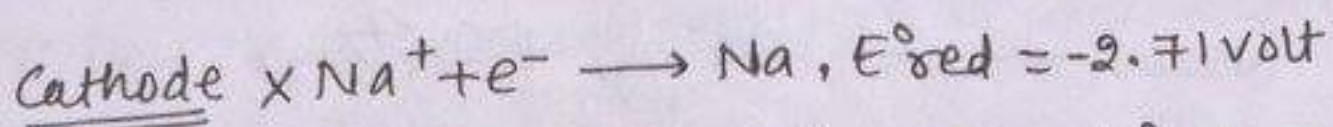
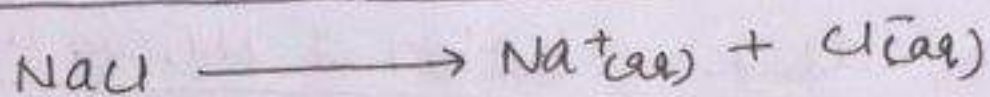
Eg OF ELECTROLYSIS (SPECIAL)

1. Electrolysis of very dil aqueous NaCl using Pt electrode
2. Electrolysis of aqueous solution NaCl using Hg electrode
3. Electrolysis of H_2SO_4 using Pt electrode
4. Electrolysis of aq. solution of CuSO_4 using Cu electrode

1. Electrolysis of fused (MOLTEN) NaCl using Pt electrode



2. Electrolysis of Aq. soln of NaCl using Pt electrode

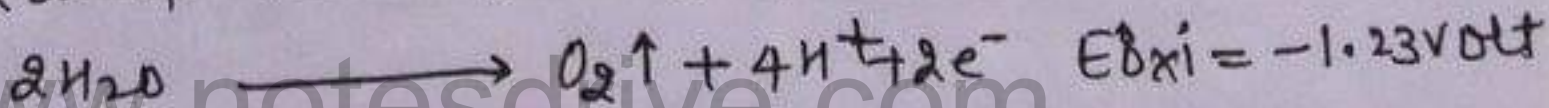
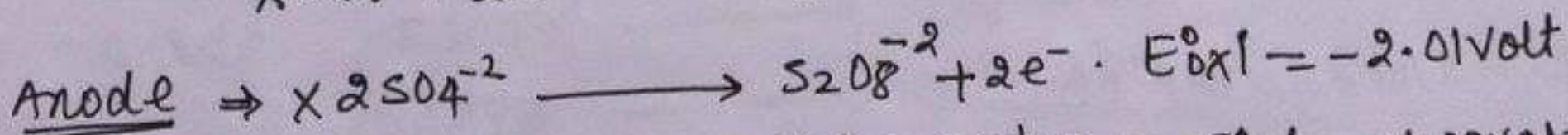
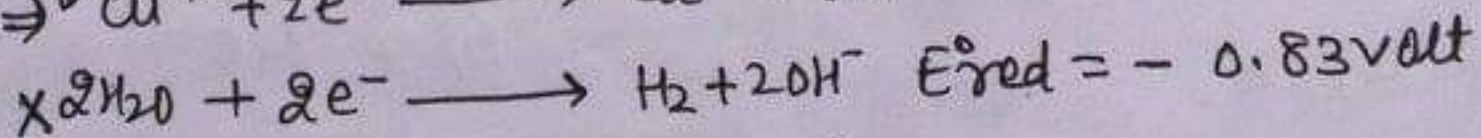
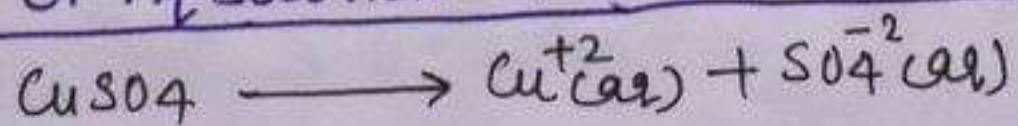


NOTE:

* S.O.P of water का Cl से ज्यादा है इसलिए water should be oxidised at anode

* But it is kinetically slow at that lower potential so some extra potential supply करना पड़ता है। इसे OVER POTENTIAL या BUBBLE POTENTIAL कहते हैं।

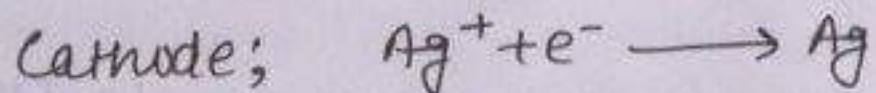
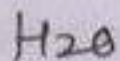
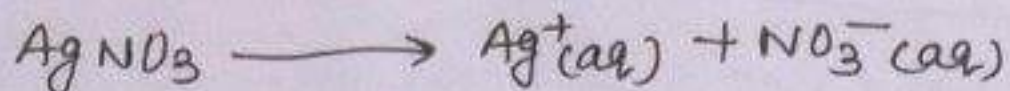
3. ELECTROLYSIS OF Aq. SOLUTION OF CuSO_4 USING Pt ELECTRODE



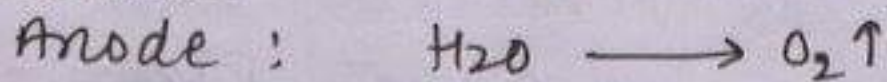
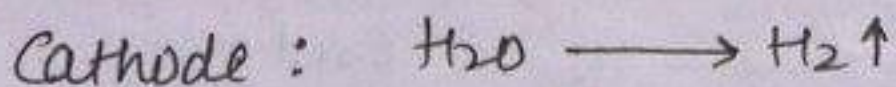
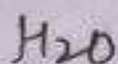
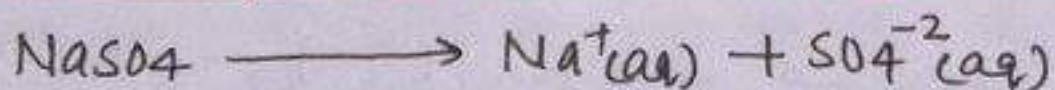
NOTE:

Cathode पर Higher SRP वाला reaction will preferred
Anode पर Higher SOP वाला reaction will preferred

4. Electrolysis of aq solution of AgNO_3 using Pt electrode

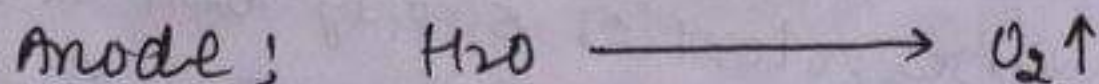
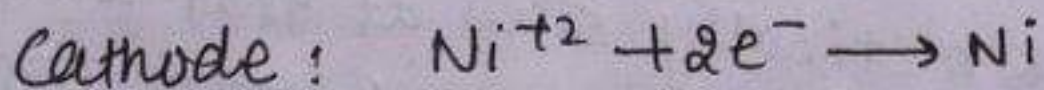
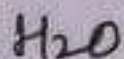
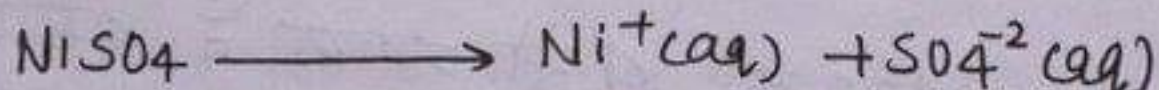


5. Electrolysis of aqueous solution of Na_2SO_4 using Pt electrode



only electrolysis of water

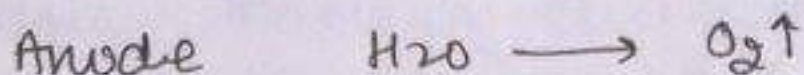
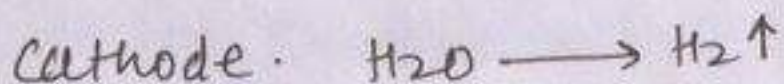
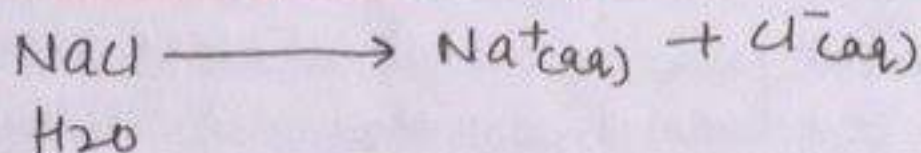
6. Electrolysis of aqueous solution of NiSO_4 using Pt electrode



KOTA CLASSNOTES

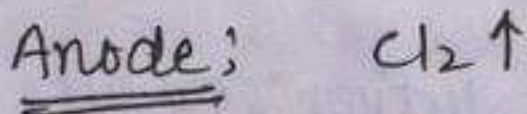
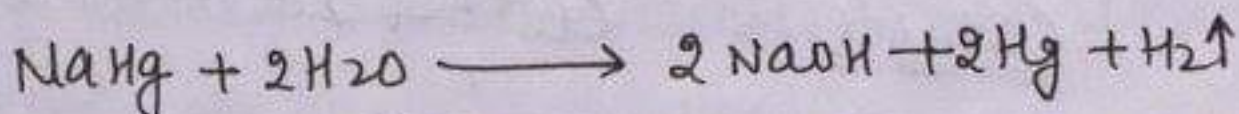
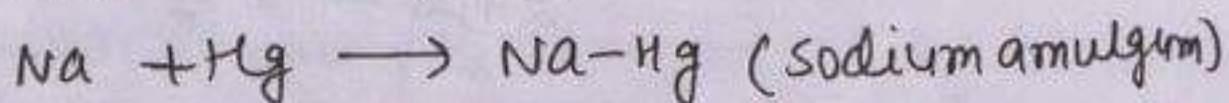
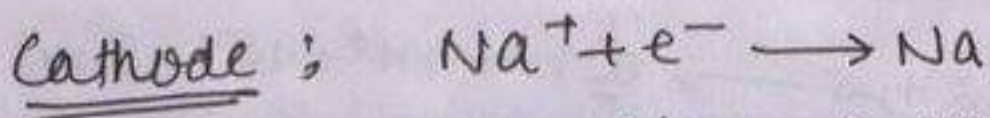
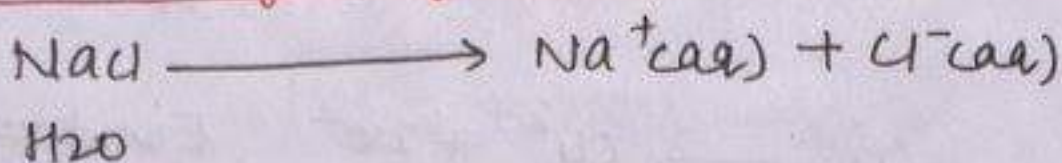
SPECIAL EXAMPLE OF ELECTROLYSIS

1. Electrolysis of very dil aq NaCl using Pt electrode



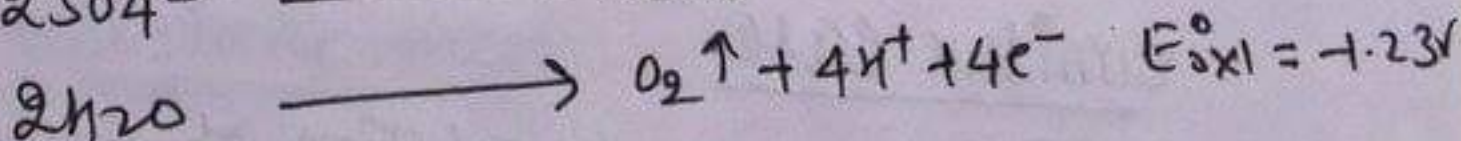
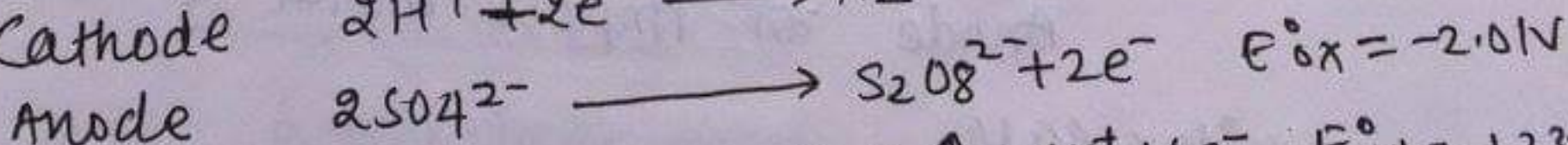
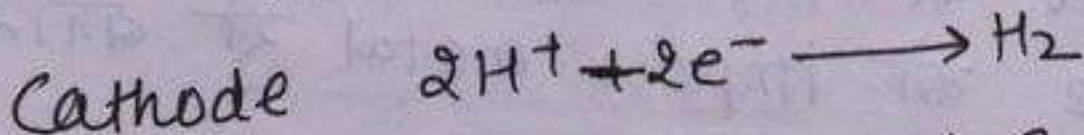
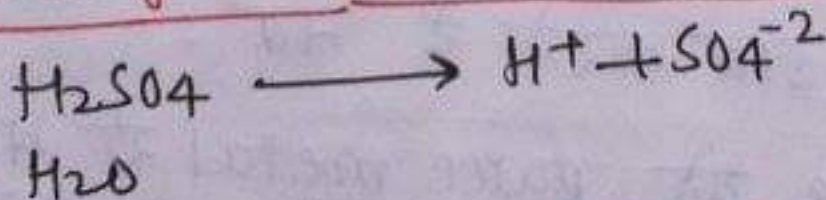
⇒ Cl^- ions के कम concentration पर discharged Potential water से ज्यादा चाहिए

2. Electrolysis of aq solution of NaCl using Hg electro



⇒ At Hg electrode discharge potential of Na^+ is lesser than H_2O

3. Electrolysis of H_2SO_4 solution using Pt electrode



Note:

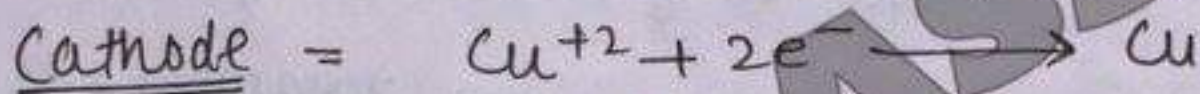
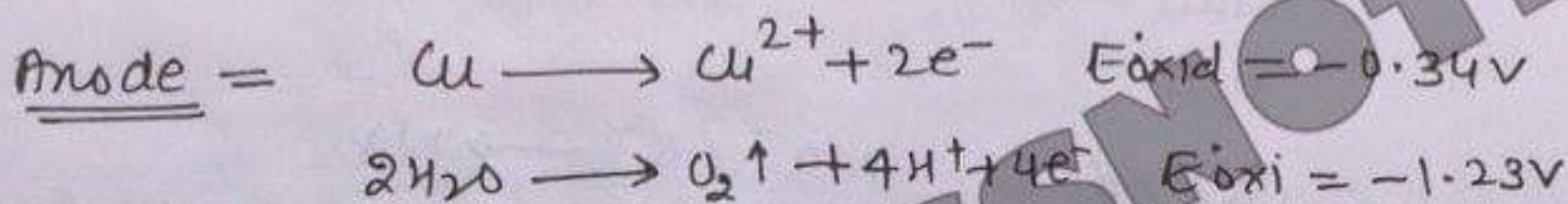
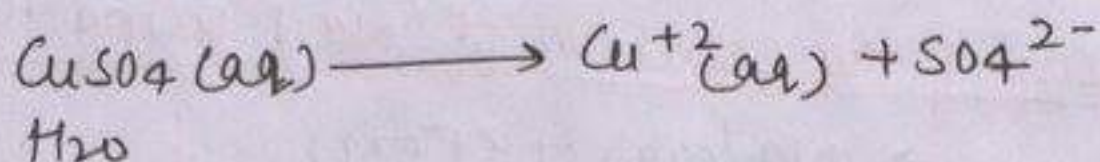
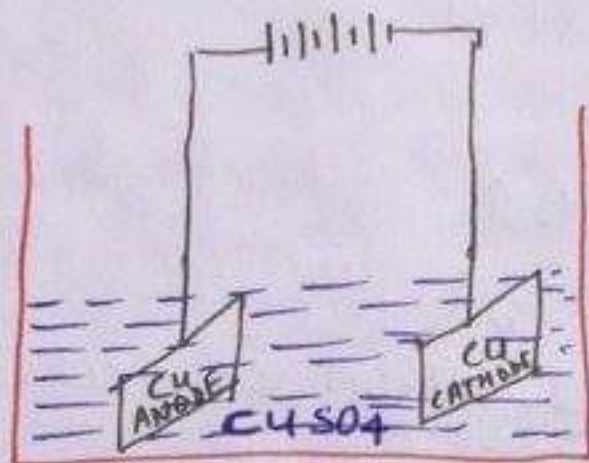
At lower concⁿ second reaction is preferred

At higher concⁿ first reaction is preferred

KOTA CLASSNOTES

www.notesdrive.com

4. Electrolysis of aqueous solution of copper sulphate using Pt electrode



NOTE

- * In this process concⁿ of solution remains same
- * It is not considered as an example of electrolysis
- * This process can be used for purification of Cu
- * Purification जब करते हैं तब :-

Cathode को pure metal से बनाते हैं।

Anode को impure metal से बनाते हैं।

Similar case

* Electrolysis of Aq solution of CuCl_2 using Cu electrode

* $\text{AgNO}_3 \longrightarrow \text{Ag}$

* $\text{Ni(NO}_3)_2 \longrightarrow \text{Ni}$

FARADAY'S LAW OF ELECTROLYSIS

⇒ FIRST LAW OF FARADAY'S LAW OF ELECTROLYSIS

When electricity is passed through electrolyte then amount of substance discharged on electrode is directly proportional to the quantity of charge passed through electrolyte

$$W \propto Q$$

$$W \propto it$$

$$\boxed{W = z \cdot it}$$

$z =$ electrochemical equivalent

If $Q = it = 1 \text{ cb}$ then

$$\boxed{W = z} \text{ gm}$$

Electrochemical equivalence

The weight of substance (in gm) discharged by 1 coulomb of charge is called electrochemical equivalence of that substance

∴ 1 Faraday charge discharges $(W) = E \text{ gm}$
(96500 cb)

∴ 1 cb charge discharge $\frac{E}{96500} = z$

$$W = z \cdot it$$

$$\boxed{W = \frac{E}{96500} \times it}$$

If $Q = it$ is constant then

$$W \propto E$$

$$\frac{W}{E} = \text{Constant}$$

⇒ SECOND LAW OF FARADAY'S LAW OF ELECTROLYSIS

if same quantity of charge is passed through different solution of electrolyte then amount of substance discharged on electrode will be in the ratio of their equivalent weight

$$\boxed{\frac{W_1}{W_2} = \frac{E_1}{E_2}} \quad \boxed{\frac{W_1}{E_1} = \frac{W_2}{E_2} = \frac{W_3}{E_3} \dots \text{Constt}}$$

Qsn Calculate the weight of Cu deposited when 2×10^3 Coulomb of charge is passed through CuSO_4 solution (Atomic weight of Cu = 63.5)

Ans $\frac{W_{\text{Cu}}}{E_{\text{Cu}}} = \frac{it}{96500}$

$$W_{\text{Cu}} = \frac{it \times E_{\text{Cu}}}{96500} = \frac{63.5 \times 2 \times 10^3}{2 \times 96500} = 0.658 \text{ gm}$$

Qsn 0.36 gm of metal is deposited on the electrode when 1.2 amp current is passed for 15 minute through its salt. Atomic weight of the metal is 96. what will be its valency

Ans $W = \frac{E \times Q}{96500} = \frac{W \times it}{\text{valency} \times 96500} = \frac{96 \times 1.2 \times 15 \times 60}{x \times 96500}$

$$0.36 = \frac{96 \times 1.2 \times 15 \times 60}{x \times 96500}$$

$$x = \frac{96 \times 1.2 \times 15 \times 60}{96500 \times 0.36}$$

$$x = 2.98$$

$$\underline{x = 3}$$

[क्योंकि x होगा एक whole number होना है।]

Qsn How much current is necessary to produce O_2 gas at the rate of 1ml/sec at STP.

Ans: $V_{O_2} = 1\text{ml}$
 $t = 1\text{sec}$

$$W_{O_2} = \frac{1}{22400} \times 32\text{ gm}$$

$$W_{O_2} = \frac{E_{O_2}}{96500} \times i t$$

Gas	MW	EW
O_2	32	8
H_2	2	1
Cl_2	71	35.5

$$\frac{32}{22400} = \frac{8}{96500} \times i \times 1\text{sec}$$

$$\underline{i = 17.23\text{ amp}}$$

Qsn How long a current of 3 amp has to be passed through a silver nitrate solution to coat a metal surface of 80 cm^2 with 0.005 mm thick layer. $\rho_{\text{silver}} = 10.5\text{ gm/cm}^3$

Ans weight of Ag deposited on metal = $v \times d$
 $= (\text{Area} \times \text{thick}) \times d$
 $= \frac{80 \times 0.005}{10} \times 10.5$
 $= 0.42\text{ gm}$

$$W_{Ag} = E_{Ag} \times \frac{i t}{96500}$$

$$0.42 = \frac{108}{96500} \times 3 \times t$$

$$\underline{t = 129.09\text{ sec}}$$

Qsn 5 lit of 1M AgNO_3 solution and 2 lit of 1M NaCl solution are connected in series when electricity is passed 54 gm of silver is deposited on cathode in AgNO_3 solution then calculate volume of H_2 gas (at STP) produced at cathode in NaCl solution ($A_g = 108$)

Ans $W = 54 \text{ gm of Ag}$

$$\frac{W_{\text{H}_2}}{W_{\text{Ag}}} = \frac{E_{\text{H}_2}}{E_{\text{Ag}}}$$

$$\frac{W_{\text{H}_2}}{54} = \frac{1}{108} \Rightarrow W_{\text{H}_2} = \frac{1}{2} = 0.5 \text{ gm}$$

$$V_{\text{H}_2} = \frac{0.5}{2} \times 22.4 = 5.6 \text{ litr}$$

$$\underline{V_{\text{H}_2} = 5.6 \text{ lit}}$$

Qsn calculate the time required to deposit 40% Cu from 200 ml of 0.5 M CuSO_4 solution when 0.2 amp current is passed through it

Ans $M \times (V)_L = \text{mole of solute}$

$$0.5 \times \frac{200}{100} = 0.1 \text{ mol of CuSO}_4$$

$$\downarrow$$

$$0.1 \text{ mol of Cu}^{2+}$$

$$\downarrow$$

$$(0.1 \times 63.5) \text{ g Cu}^{2+}$$

$$\text{wt of Cu deposited} = \frac{0.1 \times 63.5 \times 40}{100} \text{ gm}$$

$$W_{\text{Cu}} = \frac{E_{\text{Cu}} \times i \times t}{96500}$$

$$\frac{0.1 \times 63.5 \times 40}{100} = \frac{63.5 \times 0.2 \times t}{2 \times 96500}$$

$$\underline{t = 38600 \text{ sec.}}$$

Qsn If cost of electricity producing x gm of Al is Rs x . What is the cost of electricity of producing x gm of Mg (At mass of Al = 27) (At mass of Mg = 24)

Ans Cost of deposition $\propto$ charge (Q)

$$W = \frac{E}{96500} \times Q$$

$$\frac{W_{Mg}}{W_{Al}} = \frac{E_{Mg} \times Q_{Mg}}{E_{Al} \times Q_{Al}}$$

$$\frac{W_{Mg}}{W_{Al}} = \frac{E_{Mg} \times (\text{Cost})_{Mg}}{E_{Al} \times (\text{Cost})_{Al}}$$

$$\frac{x}{x} = \frac{24/2}{27/3} \times \frac{(\text{Cost})_{Mg}}{x}$$

$$1 = \frac{12}{9} \times \frac{(\text{Cost})_{Mg}}{x}$$

$$(\text{Cost})_{Mg} = x \times \frac{3}{4} = 0.75x \text{ Rs}$$

Qsn 100 gm concentration solution of $\overset{\text{Co}}{\text{AgNO}_3}$ is electrolysed using 0.1 Faraday of electricity by Pt electrodes. Find the weight of resulting solution (At mass of Ag = 108)

Ans $1F \longrightarrow W = E_{gm}$

$$0.1F \longrightarrow W = (0.1 \times E) = 0.1 \times 108$$

$$\text{Cathode} \longrightarrow W_{Ag} = 10.8 \text{ gm}$$

$$\text{Anode} \longrightarrow W_{O_2} = 0.1 \times 8 = 0.8 \text{ gm}$$

$$\begin{aligned} \text{Remaining wt of solution} &= 100 - (10.8 + 0.8) \\ &= 88.4 \text{ gm} \end{aligned}$$

Qsn A current of 3 Amp was passed for 2 hours through a solution of CuSO_4 . 3 gm of Cu^{+2} ions were discharged at cathode. Calculate current efficiency for this process ($\text{Cu} = 63.5$)

Ans $W = \frac{E \times i \times t}{96500}$

$$3 = \frac{63.5}{2 \times 96500} \times i_{\text{used}} \times 2 \times 60 \times 60$$

$$i_{\text{used}} = 1.266 \text{ Amp}$$

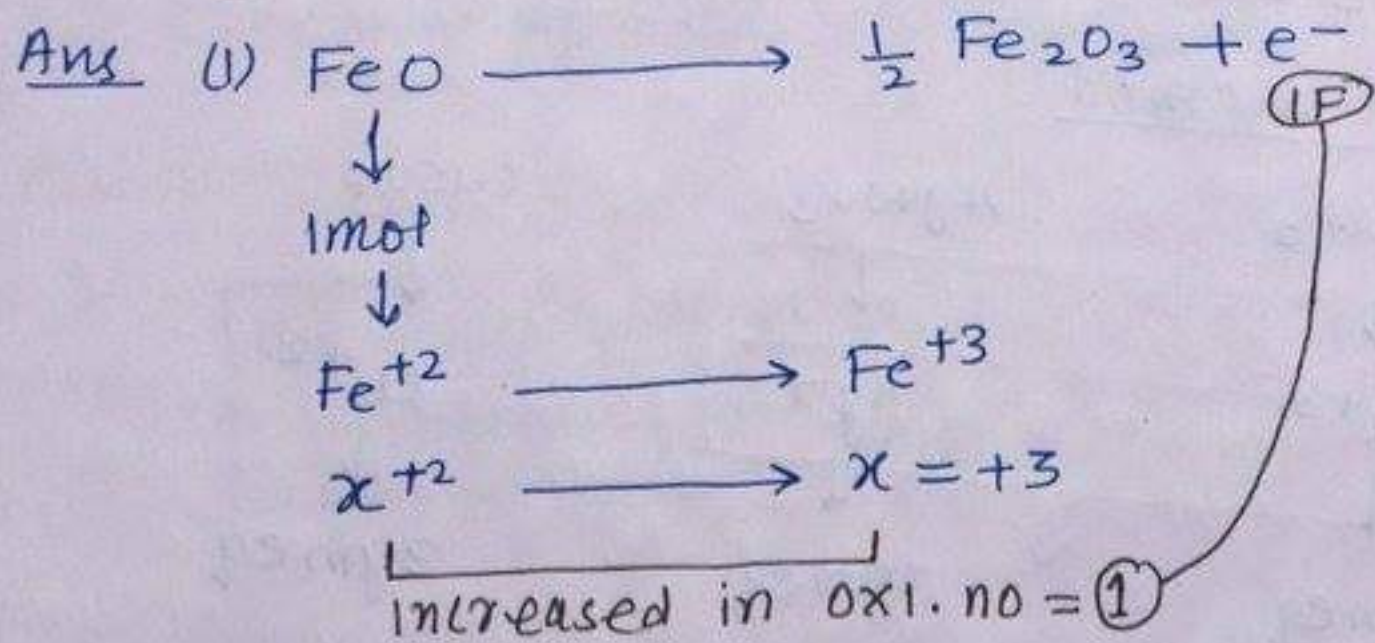
$$\text{Current efficiency} = \frac{i_{\text{used}}}{i_{\text{passed}}} \times 100$$

$$= \frac{1.266}{3} \times 100$$

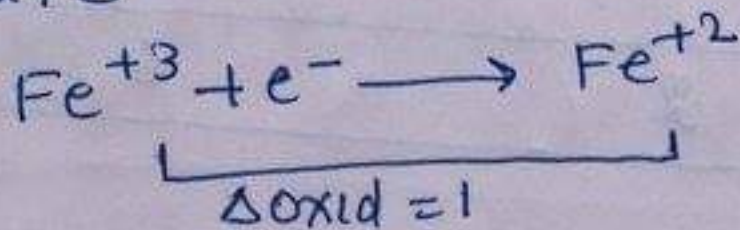
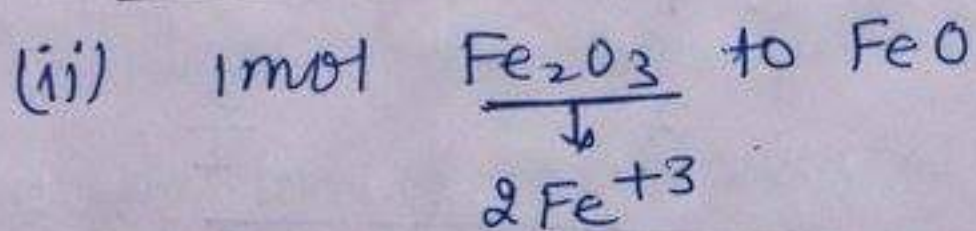
$$= 42.2 \%$$

KOTA CLASSNOTES

- Qsn Calculate charge require for oxidation of
- 1 mol FeO to Fe_2O_3
 - 1 mol Fe_2O_3 to FeO
 - 1 mol H_2O to O_2

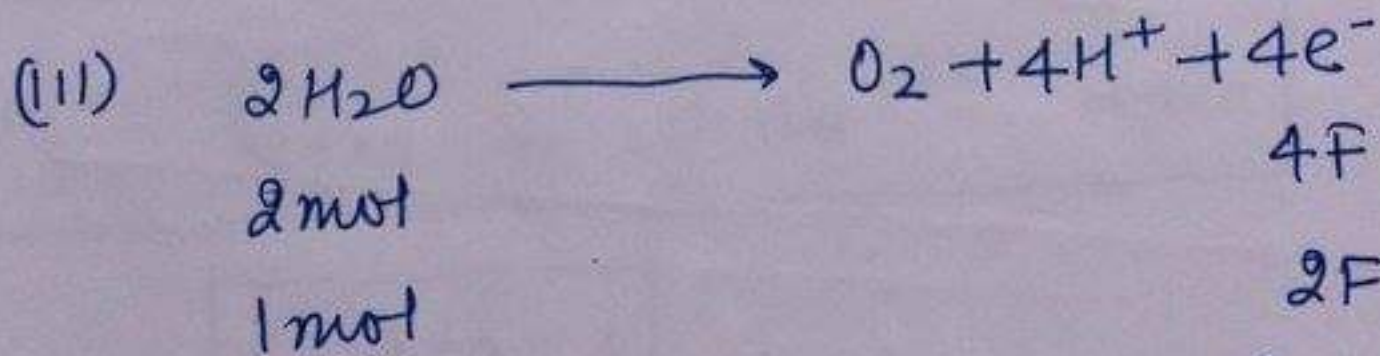


Ans = 1F , $Q = 96500 \text{ Cb}$



1 mol = 1 Faraday

2 mol = 2 Faraday

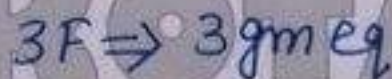
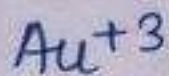
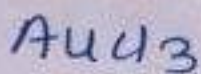


$Q = 2 \times 96500$

$= 193000 \text{ Cb}$

Q57 Aqueous solution of AuCl_3 , AgNO_3 , CuSO_4 are connected in series when 3F charge is passed then calculate the ratio of moles of metal deposited on electrode

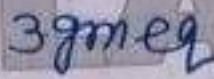
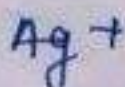
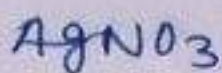
Ans 1st method



$$v.f \Rightarrow \frac{3\text{mol}}{3}$$

$$v.f \Rightarrow 1$$

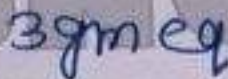
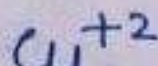
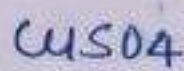
$$v.f \Rightarrow 2$$



$$\frac{3\text{mol}}{1}$$

$$3$$

$$6$$



$$\frac{3\text{mol}}{2}$$

$$\frac{3}{2}$$

$$3$$

Relation between ionic Conductance (χ) and mobility of ion (u)

$\chi \propto u$
for cation $\chi_c \propto u_c$
for Anion $\chi_a \propto u_a$

$$\chi_c = F u_c$$
$$\chi_a = F u_a$$

$$F = 96500 \text{ J/mol.}$$

mobility of ion (u).

$$u = \frac{\text{Velocity of ion (v)}}{\text{potential gradient}}$$



transport No(t).

- ★ The current flowing through an electrolytic solution is carried by the ions (Cation anion) both.
- * The fraction of total current carried by an anion is called its transport number

$$t_c = \frac{\lambda_c}{\lambda_c + \lambda_a} \quad \text{or} \quad \frac{u_c}{u_c + u_a} \quad \text{or} \quad \frac{V_c}{V_c + V_a}$$

$$t_a = \frac{\lambda_a}{\lambda_c + \lambda_a} \quad \text{or} \quad \frac{u_a}{u_c + u_a} \quad \text{or} \quad \frac{V_a}{V_c + V_a}$$

$$t_c + t_a = 1$$

GALVANIC CELL	ELECTROLYTIC CELL
* Convert chemical energy into electrical energy	* Converts electrical energy into chemical energy
* Cell reaction is spontaneous	* cell reaction is Non-spontaneous
* Anode = -ve Cathode = +ve	* Anode = +ve cathode = -ve
* Deposition of substances only at cathode	* deposition of substance on both electrode cathode and anode
* Salt Bridge is used	* salt Bridge is not used
* Both electrodes are separated	* Both electrodes are placed in same compartment
* Current flows from cathode to anode	* current flows from anode to cathode

BATTERY	ANODE	CATHODE	ELECTROLYTE
DRY CELL / PRI CELL (NON-RECHARGEABLE)	Zn	Carbon Rod Coated with MnO_2	NH_4Cl with $ZnCl_2$
Lead storage Battery	Pb	Pb rod Coated with PbO_2	38% By mass of H_2SO_4
Ni-cd cell (secondary cell) Rechargeable	Cd	Ni rod Coated with Nickel Powder	Aqueous solution of $KOH/NaOH$
H_2O_2 Fuel cell Pollution free cell	H_2 gas passing through Porous Carbon	O_2 passing through Porous Carbon	Aqueous solution of $KOH/NaOH$

RA